Online Appendix: The Real Cost of Benchmarking*

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B Additional Results

B.1 Additional Tables

Table B1: Robustness to Alternative CAPM $\hat{\beta}$ Estimators

	(1) $\Delta\ \widehat{\beta}^{OLS}$	(2) $\Delta\ \widehat{\beta}^{WEL}$	(3) $\Delta\ \widehat{\beta}^{DIM}$	(4) $\Delta\ \widehat{\beta}^{BLU}$	(5) $\Delta\ \rho(r_i, r_m)$	(6) $\Delta\ \sigma^i$
Δ BMI (in p.p.)	0.0183***	0.0154***	0.0135***	0.0122***	0.00646***	-0.0000668*
	(0.001)	(0.001)	(0.002)	(0.001)	(0.000)	(0.000)
Firm Fixed Effects	✓	✓	✓	✓	✓	✓
Year Fixed Effects	✓	✓	✓	✓	✓	✓
Adj. R^2	0.15	0.16	0.11	0.15	0.41	0.59
Observations	28,514	28,514	28,514	28,514	28,514	28,514

Notes: This table reports coefficient estimates of specifications of the form: $\Delta\hat{\beta}_{i,t} = \alpha_i + \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \varepsilon_{i,t}$. $\Delta\beta_{i,t}$ is between 1st and 4th quarter of each year. β^{WEL} is estimator of Welch (2022), β^{DIM} is estimator of Dimson (1979), β^{BLU} is estimator of Blume (1975) (also known as Bloomberg $\hat{\beta}$). We winsorize $\hat{\beta}$s changes at the 2% and 98% level. We restrict the estimation sample to stocks within 300 ranks around Russell index 1000/2000 and 3000 cutoffs. Standard errors in parentheses are double clustered at firm- and year level. * p<0.10, ** p<0.05, *** p<0.01.

Table B2: Effects of Benchmarking Intensity on Equal-Weighted Average CAPM $\hat{\beta}^E$

	(1) OLS	(2) ADL(1)	(3) DOLS(3)	(4) DOLS(4)	(5) DOLS(5)	(6) DOLS(7)
Average Benchmarking Intensity (in %)	0.038	0.030*	0.032***	0.031***	0.031***	0.029***
		(0.015)	(0.005)	(0.005)	(0.005)	(0.005)
Engle-Granger's Augmented Dickey-Fuller test (H_0: no cointegration)		-13.77***	-3.93***	-3.66**	-3.75**	-3.30*
Adj. R^2		0.99	0.71	0.72	0.74	0.78
Observations	225	225	225	225	225	225

Notes: This table estimates the long-run relationship between the equal-weighted average CAPM $\hat{\beta}$ and the average Benchmarking Intensity (BMI). To address the potential for spurious regression in the time series, we report results from several models. Column (1) shows the static OLS estimate. Column (2) presents an Autoregressive Distributed Lag, ADL(1), model. Columns (3)-(6) report long-run coefficients from Dynamic OLS (DOLS) (Stock & Watson, 1993) specifications with varying numbers of leads and lags. We report the Engle-Granger Augmented Dickey-Fuller test statistic for the residuals of the cointegrating regression, where the null hypothesis is no cointegration. Newey-West standard errors (21 lags) in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table B3: Panel Regression of Firm Investment Accounting for Measurement Error in Tobin's q

	(1)	(2)	(3)	(4)	(5)
	Dep. var.: Investment rate (CAPX/PPE)				
	OLS		IV: Industry q		EW (2012)
BMI (in %)	-0.147***	-0.105***	-0.099***	-0.070***	-0.067***
	(0.021)	(0.022)	(0.022)	(0.020)	(0.014)
Tobin's q^{tot}	3.434***	3.038***	5.043***	4.573***	6.078***
	(0.155)	(0.205)	(0.327)	(0.435)	(0.157)
Leverage	0.008*	0.012**	0.018***	0.006	0.001
	(0.005)	(0.006)	(0.006)	(0.006)	(0.005)
Firm FE		✓		✓	✓
Year × Rank-Bin FE	✓	✓	✓	✓	✓
FS F-stat.			143.16	33.95	
Within R^2	0.21	0.12	0.17	0.10	0.27
Observations	36,277	36,129	35,690	35,543	36,129

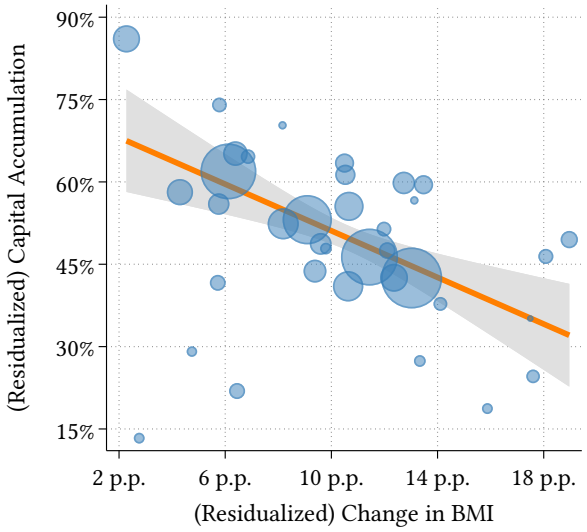
Notes: This table report estimates for panel regressions of the form: $\frac{\text{CAPX}_{i,t+1}}{\text{PPE}_{i,t}} = \alpha_{t,bin} + \alpha_i + \gamma\,\text{BMI}_{i,t} + X'_{i,t}\xi + \varepsilon_{i,t}$, where $\alpha_{t,bin}$ is a year-by-rank-bin fixed effect with bins defined every 250 market capitalization ranks in May. We address measurement error in Tobin's q (Peters & Taylor, 2017) in columns 3-5. Columns 3-4 use the NAICS-4 industry level leave-one-out mean of Tobin's q as an instrument for firm-level Tobin's q. Columns 5 use Erickson & Whited's (2012) cumulant estimator. We additionally control for leverage and cash flow to PPE. Standard errors in parentheses are clustered at the year- and firm-level. * p<0.10, ** p<0.05, *** p<0.01.

Table B7: BEA Capital Accumulation vs. Change in BMI

	(1)	(2)	(3)	(4)
	Dep. var.: Capital Accumulation (1998-2018)			
Change in BMI	-1.683**	-2.010***	-1.960**	-1.660**
(1998-2018)	(0.703)	(0.624)	(0.757)	(0.787)
Capital/Value-Added			-0.008	-0.004
(1998)			(0.007)	(0.007)
Log(Value-Added)			0.006	0.023
(1998)			(0.031)	(0.034)
Log(Labor)			0.048	0.028
(1998)			(0.049)	(0.046)
Pre-trend Capital				0.115
(1980-1996)				(0.087)
NAICS-2 Fixed Effect		✓	✓	✓
Weights		VA_{1998}	VA_{1998}	VA_{1998}
Adj. R^2	0.10	0.70	0.70	0.71
Observations	38	38	38	38

Notes: This table shows estimates of regression of the form $\log\left(\text{Real Capital}_i^{'18}/\text{Real Capital}_i^{'98}\right) = \alpha_j + \gamma\Delta_{98}^{18}\,\text{BMI}_i + X'_i\zeta + \varepsilon_i$ for industry i in sector j from 1998 to 2018 using the BEA Fixed Asset Table. We weight each industry by its value-added share in 1998. Robust standard errors in parentheses.

Figure B1: Capital Accumulation vs. Change in BMI



Notes: This figure plots residualized capital accumulation versus residualized changes in benchmarking intensity for 3-digit NAICS industries in long-differences from 1998 to 2018 using the BEA Fixed Asset Table. The specification is the same as in Column (3) of Table B7. Pointwise confidence interval (95%) based on robust standard errors.

Table B4: Panel Regressions of Firm Investment on BMI (without imposing sample restrictions)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: CAPX_{t+1} / PPE_t (in %)						
BMI (in %)	-0.214***	-0.349***	-0.191***	-0.114***	-0.124***	-0.134***	-0.186***
	(0.044)	(0.033)	(0.033)	(0.020)	(0.022)	(0.027)	(0.036)
IOR (in %)				0.006	0.004	0.004	0.003
				(0.005)	(0.006)	(0.006)	(0.006)
Tobin's q^{tot}				2.192***	2.092***	2.087***	
				(0.145)	(0.142)	(0.142)	
Linear Time Trend	-0.423***						
	(0.071)						
Leverage				0.019***	0.018***	0.017***	0.022***
				(0.005)	(0.005)	(0.005)	(0.006)
Fixed Effects							
Firm FE	✓		✓	✓	✓	✓	✓
Year × Rank FE		✓	✓	✓			
Ind. × Year × Rank FE					✓	✓	✓
Russell 2000 Index FE						✓	✓
Adj. R^2	0.43	0.07	0.46	0.49	0.56	0.56	0.53
Mean Dep. Var.	14.6	14.7	14.6	14.6	14.6	14.6	14.6
SD BMI	9.2	9.2	9.2	9.1	9.1	9.1	9.1
Observations	53,017	53,763	53,015	50,796	49,553	49,553	49,567

Notes: In this table, we do not impose any sample restrictions on firm size, industry, or data availability. This table report estimates for panel regressions of the form: $\frac{\text{CAPX}_{i,t+1}}{\text{PPE}_{i,t}} = \alpha_{t,bin} + \alpha_i + \gamma\,\text{BMI}_{i,t} + X'_{i,t}\xi + \varepsilon_{i,t}$, where $\alpha_{t,bin}$ is a year-by-rank-bin fixed effect with bins defined every 250 market capitalization ranks in May. Controls include institutional ownership ratio (IOR), Tobin's q (Peters & Taylor, 2017), leverage, cash flow to PPE, current ratio, log of market capitalization, and firm age. Standard errors in parentheses are clustered at the year- and firm-level. * p<0.10, ** p<0.05, *** p<0.01.

Table B5: Reduced Form of Russell Reconstitution BMI Shock on Investment Rate (CAPX/PPE) - Robustness to Window Size

Panel A. Window around Russell 1000/2000 Cutoff = 150						
	(1) t+1	(2) t+2	(3) t+3	(4) t+4	(5) t+5	(6) t+6
Δ BMI	0.134	-0.394	-0.477	-0.815	-0.852	-1.219**
	(0.271)	(0.390)	(0.381)	(0.496)	(0.541)	(0.524)
Window	150	150	150	150	150	150
Adj. R^2	0.07	0.10	0.09	0.09	0.09	0.07
Observations	5,058	4,826	4,342	3,896	3,458	3,072
Panel B. Window around Russell 1000/2000 Cutoff = 250						
	(1) t+1	(2) t+2	(3) t+3	(4) t+4	(5) t+5	(6) t+6
Δ BMI	0.000	-0.490	-0.555	-0.970**	-1.016**	-0.809*
	(0.278)	(0.354)	(0.354)	(0.454)	(0.430)	(0.392)
Window	250	250	250	250	250	250
Adj. R^2	0.07	0.10	0.09	0.09	0.09	0.07
Observations	7,702	7,301	6,570	5,885	5,230	4,657
Panel C. Window around Russell 1000/2000 Cutoff = 350						
	(1) t+1	(2) t+2	(3) t+3	(4) t+4	(5) t+5	(6) t+6
Δ BMI	0.105	-0.338	-0.762*	-1.006**	-1.183**	-0.817*
	(0.263)	(0.384)	(0.388)	(0.466)	(0.488)	(0.449)
Window	350	350	350	350	350	350
Adj. R^2	0.06	0.09	0.10	0.10	0.09	0.07
Observations	9,851	9,344	8,429	7,560	6,770	6,044
Panel D. Window around Russell 1000/2000 Cutoff = 450						
	(1) t+1	(2) t+2	(3) t+3	(4) t+4	(5) t+5	(6) t+6
Δ BMI	0.203	-0.227	-0.738*	-0.898*	-1.077**	-0.590
	(0.257)	(0.368)	(0.416)	(0.500)	(0.493)	(0.394)
Window	450	450	450	450	450	450
Adj. R^2	0.06	0.10	0.10	0.10	0.08	0.06
Observations	12,067	11,456	10,342	9,308	8,346	7,472

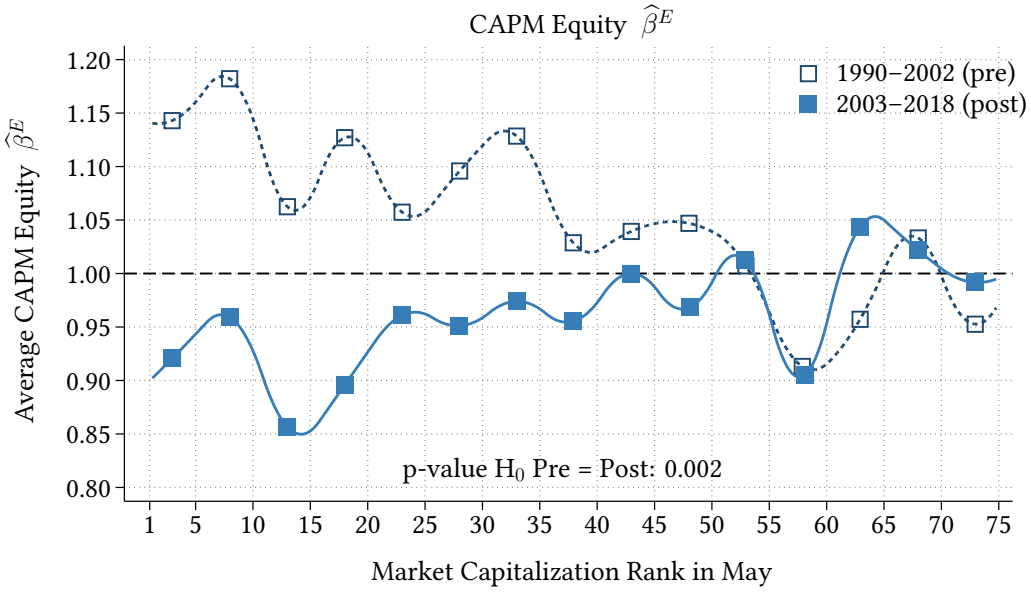
Notes: This table reports estimates for regressions of the form: $\log\left(\frac{\text{CAPX}_{i,t+h}}{\text{PPE}_{i,t}}\right) - \log\left(\frac{\text{CAPX}_{i,t}}{\text{PPE}_{i,t-1}}\right) = \alpha_t + \gamma\Delta\ \text{BMI}_{i,t} + \log\left(\text{Market Cap.}\right)_{i,t} + \text{Banding Variables}_{i,t} + \varepsilon_{i,t+h}$ Standard errors in parentheses are clustered at the year- and firm-level. * p<0.10, ** p<0.05, *** p<0.01.

Table B6: Reduced Form Effects of Benchmarking on Capital Accumulation at the Industry-level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: log (Real Capital Stock in 2016/Real Capital Stock in 1998)						
Δ CAPM β	-0.258**	-0.140*			-0.132*		
(1998-2016)	(0.111)	(0.072)			(0.078)		
Δ BMI			-0.653**			-0.608**	
(1998-2016)			(0.259)			(0.269)	
Δ IOR				0.0632			0.0931
(1998-2016)				(0.111)			(0.107)
Capital Stock/Value-Added		-0.170***	-0.185***	-0.180***	-0.221***	-0.230***	-0.224***
(1998)		(0.042)	(0.035)	(0.035)	(0.052)	(0.050)	(0.048)
log (Employment)		0.00976	0.00936	0.0188	0.0536	0.0573	0.0492
(1998)		(0.029)	(0.029)	(0.028)	(0.035)	(0.035)	(0.034)
log (TFP)		0.372	0.448*	0.397	-0.119	-0.0442	-0.104
(1998)		(0.226)	(0.230)	(0.274)	(0.228)	(0.236)	(0.224)
Pre-trend Capital					-0.217**	-0.163*	-0.137
(1980-1996)					(0.094)	(0.089)	(0.102)
Pre-trend Employment					0.195*	0.158	0.155
(1980-1996)					(0.106)	(0.096)	(0.106)
Pre-trend Wage					0.664	0.494	0.769*
(1980-1996)					(0.399)	(0.405)	(0.407)
Constant	0.238***	0.308**	0.323**	0.211			
	(0.063)	(0.150)	(0.152)	(0.143)			
NAICS-3 Subsector FE					✓	✓	✓
Weights	VA_{1998}	VA_{1998}	VA_{1998}	VA_{1998}	VA_{1998}	VA_{1998}	VA_{1998}
Adj. R^2	0.10	0.34	0.35	0.31	0.52	0.52	0.50
Observations	103	103	103	103	103	103	103

Notes: This table reports coefficient estimates of regressions at the NAICS 5-digit industry-level of the form: $\Delta \log(\text{Real Capital Stock})_i = \alpha_j + \gamma \Delta Z_i + X_i'\xi + \varepsilon_i$ for changes from 1998 to 2016. Z is either CAPM $\hat{\beta}$, benchmarking intensity (BMI), or institutional ownership ratio (IOR). BMI, IOR, and CAPM $\hat{\beta}$ are market-value weighted averages at the industry level of Compustat firms. We exclude industries with less than 5 firms. Pre-trends measure log changes in variables from 1980 to 1996. We weight observations by industry value-added in 1998 and winsorize all variables at the 2% and 98% level. Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

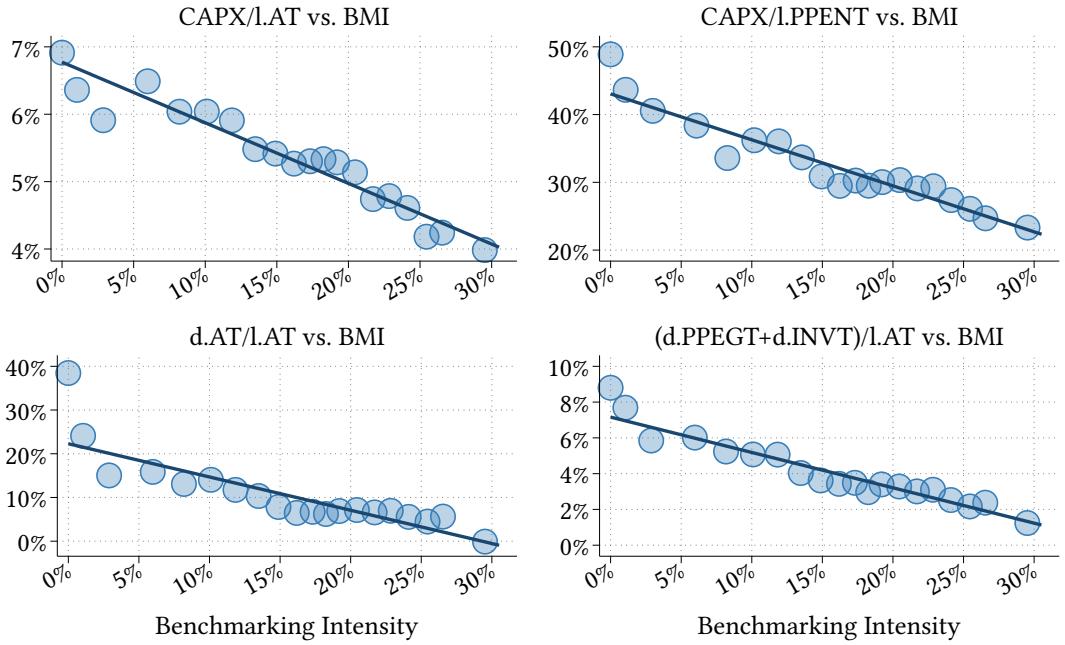
Figure B2: Largest Stocks' CAPM $\hat{\beta}^E$ vs. Market Capitalization Rank in May



Notes: This figure shows binned scatter plots of CAPM equity $\hat{\beta}^E$ against May market capitalization ranks for the largest 100 stocks by market capitalization. Each bin reflects the equal-weighted average of 5 ranks. Conditional means are identified absorbing year-month and stock fixed effects. Outlined bins use data from 1990 to 2002; filled bins from 2003 to 2018. p-value is for the null hypothesis that $\hat{\beta}(\text{rank bin } j)_{t=Pre} = \hat{\beta}(\text{rank bin } j)_{t=Post}\ \forall\ j \in \mathcal{J}$ using the nonparametric pairwise group comparison test of Cattaneo et al. (2024).

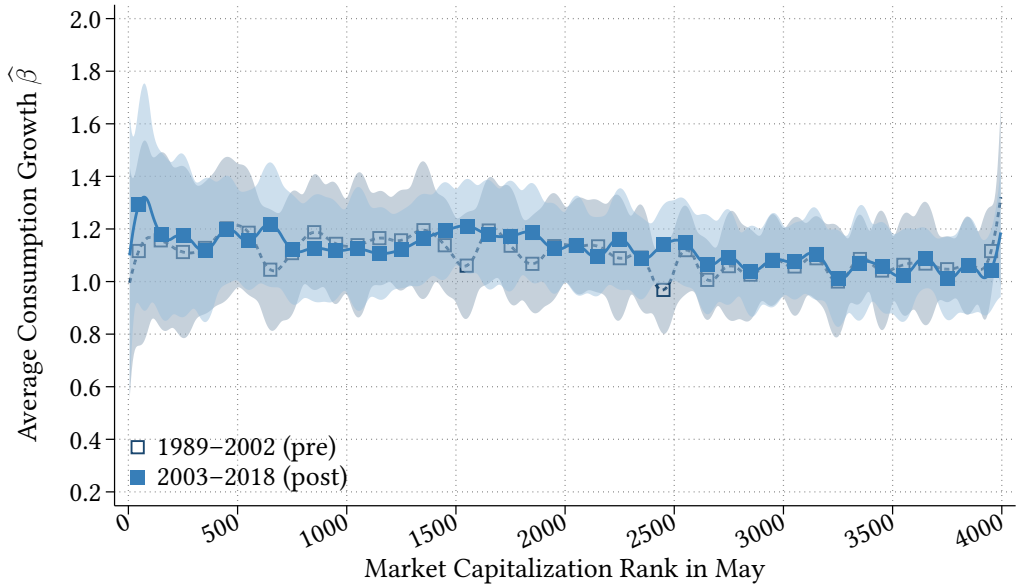
B.2 Additional Figures

Figure B3: Alternative Investment Rates Definitions



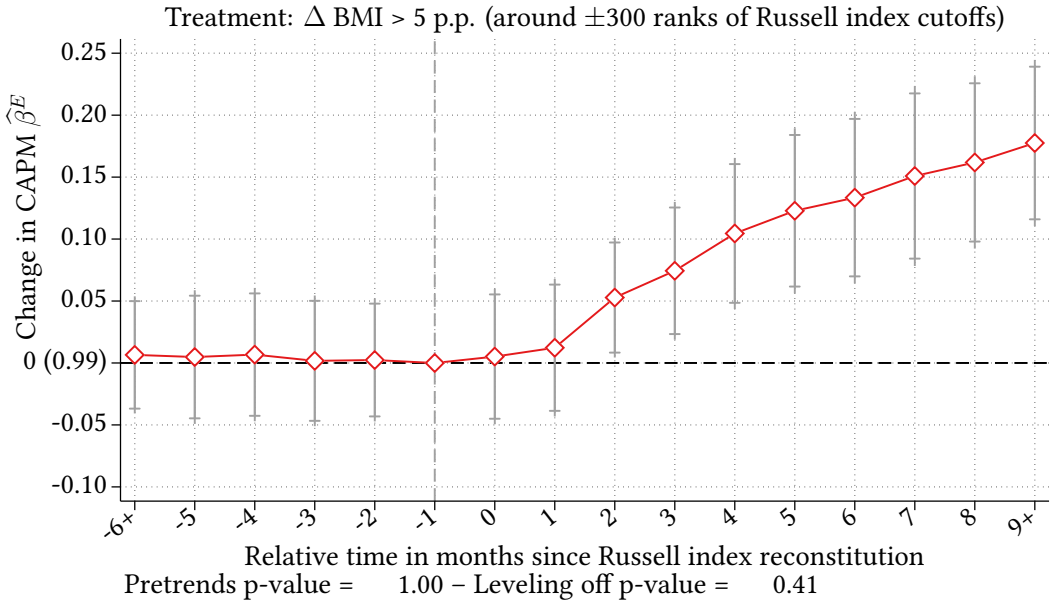
Notes: This figure plots binned scatters of 25 quantile-spaced bins of different investment rates against benchmarking intensity, constructed after absorbing year-by-rank-bucket fixed effects, with buckets defined every 250 market capitalization ranks. Sample period from 1998 2018 sample for which BMI is available.

Figure B4: Consumption Growth $\hat{\beta}$s vs. Market Capitalization Rank in May

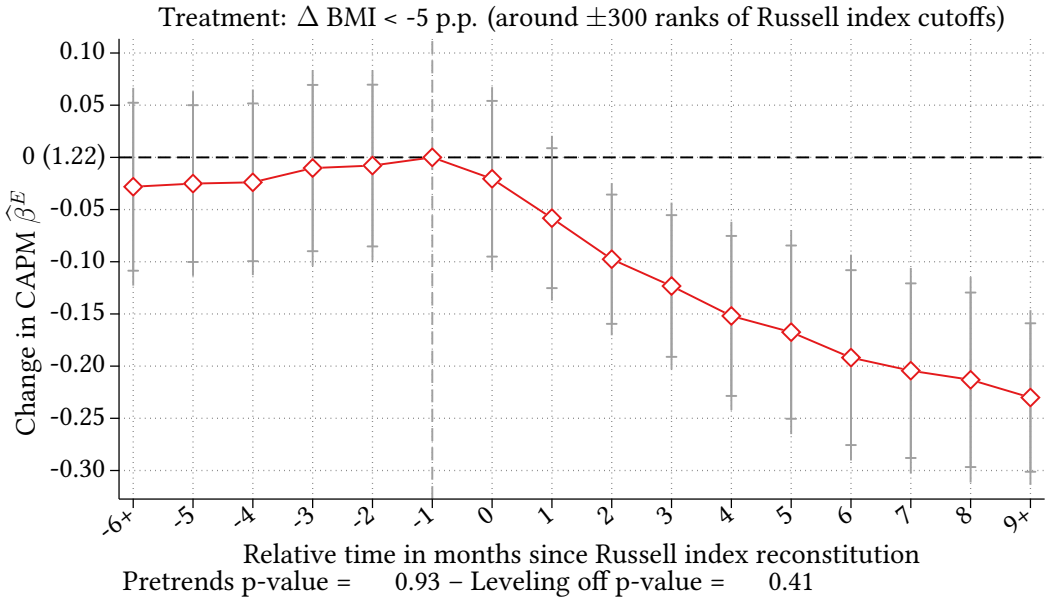


Notes: This figure shows binned scatter plots of consumption growth $\hat{\beta}$s from Kim et al. (2024) against May market capitalization ranks. Each bin reflects the equal-weighted average of approx. 100 ranks. We standardize the consumption growth $\hat{\beta}$s to have a mean and a standard deviation of one across the full sample period. Conditional means are identified absorbing yearly effects. Outlined bins use data from 1989 to 2002; filled bins from 2003 to 2018. Shaded areas show 95% confidence bands with standard errors clustered by stock and year-month.

Figure B5: Difference-in-differences Event Study of Changes in BMI on Changes in CAPM $\hat{\beta}^E$



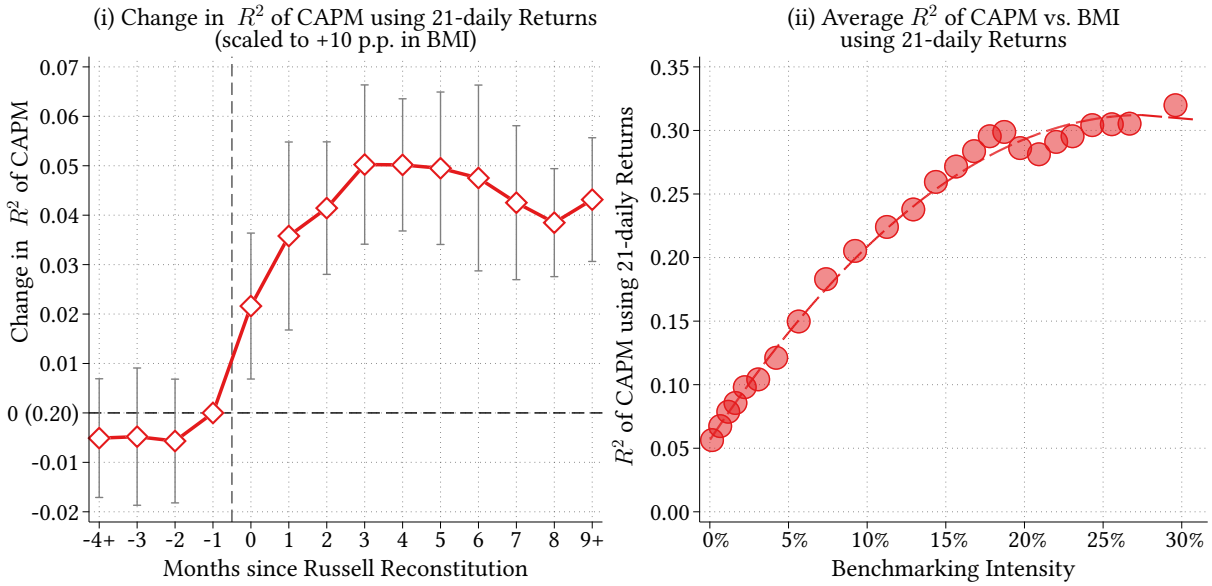
(a) Treated stocks with an increase in Δ BMI > 5 p.p. relative to control group.



(b) Treated stocks with a decrease in Δ BMI < -5 p.p. relative to control group.

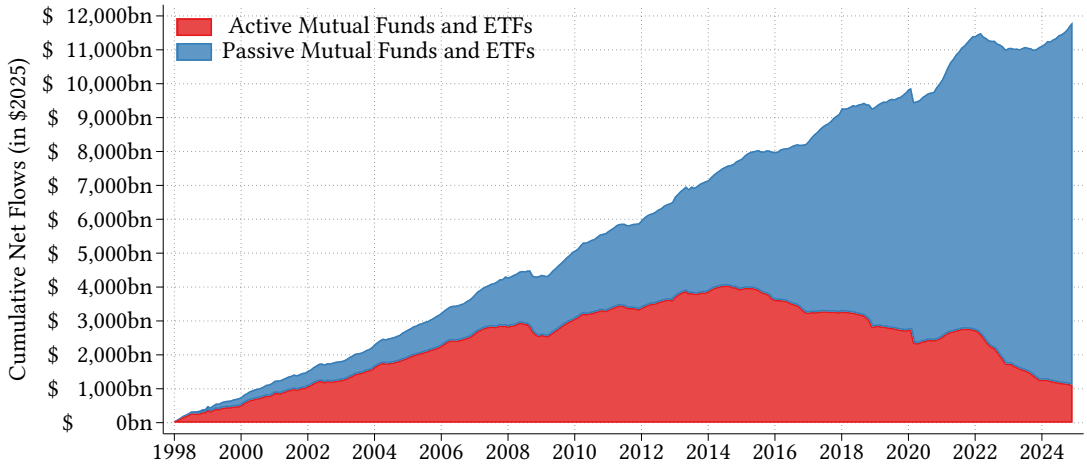
Notes: This figure shows difference-in-differences event study coefficients for a change in CAPM $\hat{\beta}^E$ of treated stocks with an increase or decrease in BMI of at least 5 p.p. relative to a control group, in which $|\Delta \text{ BMI}| < 1$ p.p.. Pointwise confidence intervals (95%) and sup-t confidence bands based on double-clustered standard errors. Values in parentheses on the Y-axis show the average CAPM $\hat{\beta}^E$ before treatment.

Figure B6: Effects of Benchmarking Intensity on the R^2 of the CAPM



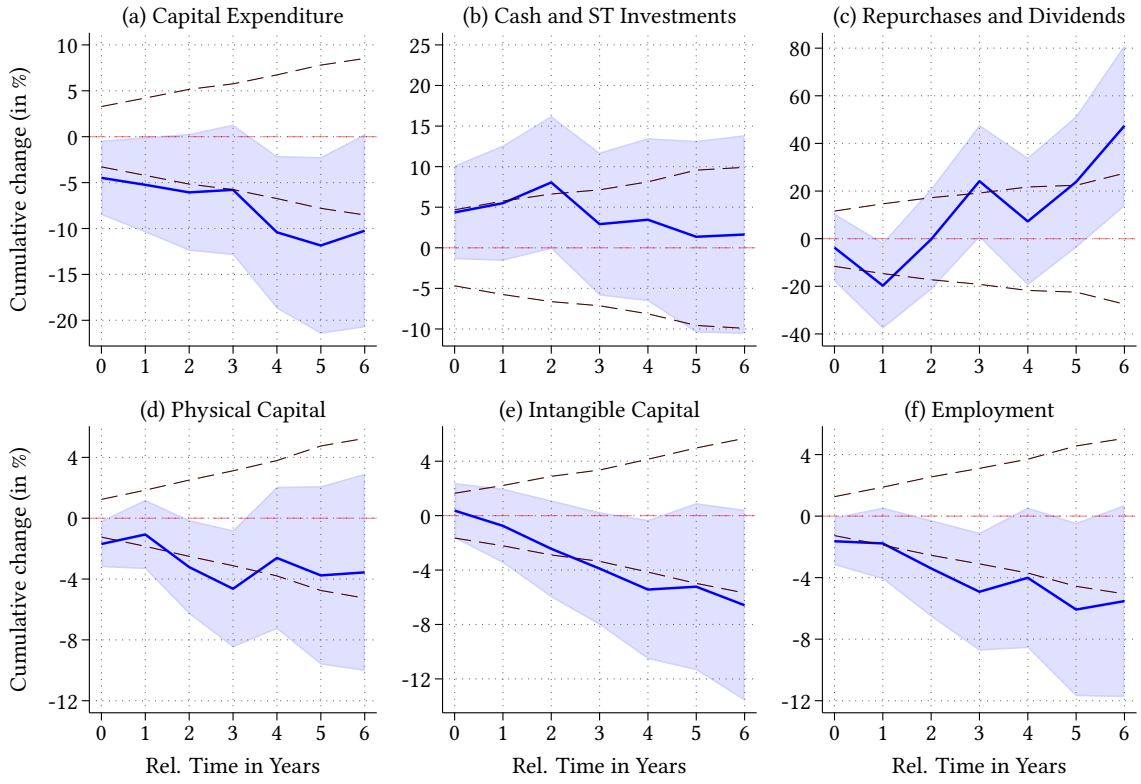
Notes: Panel (i) in this figure shows the results of a difference-in-differences event study of a 10 p.p. increase in BMI due to Russell reconstitution on the R^2 of the CAPM in explaining realized daily returns. We restrict the sample to stocks within 300 ranks of each Russell cutoff. Pointwise confidence intervals (95%) based on double-clustered standard errors. Panel (ii) shows 25 quantile-spaced bins, constructed after absorbing year-month-by-FF49-industry fixed effects, of the monthly R^2 of the CAPM versus BMI.

Figure B7: Cumulative Net Flows into Passive and Active Mutual Funds and ETFs



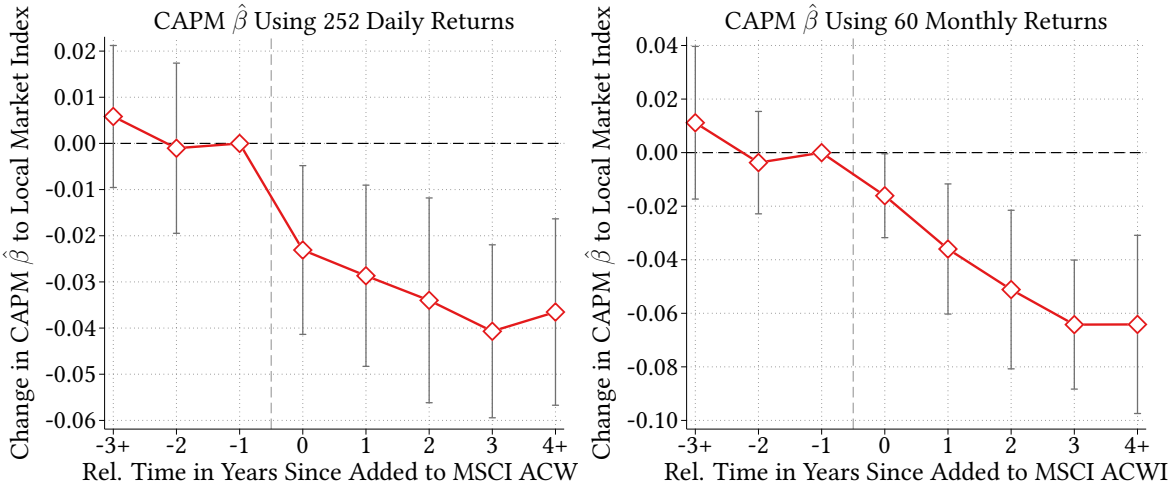
Notes: This figure show cumulative monthly net flows into active and passive mutual funds and ETFs from 1998 to 2024 deflated by the Consumer Price Index. Source: Morningstar Direct.

Figure B8: LP-IV: Impulse Response of Outcome to 100 bps CAPM-implied WACC shock



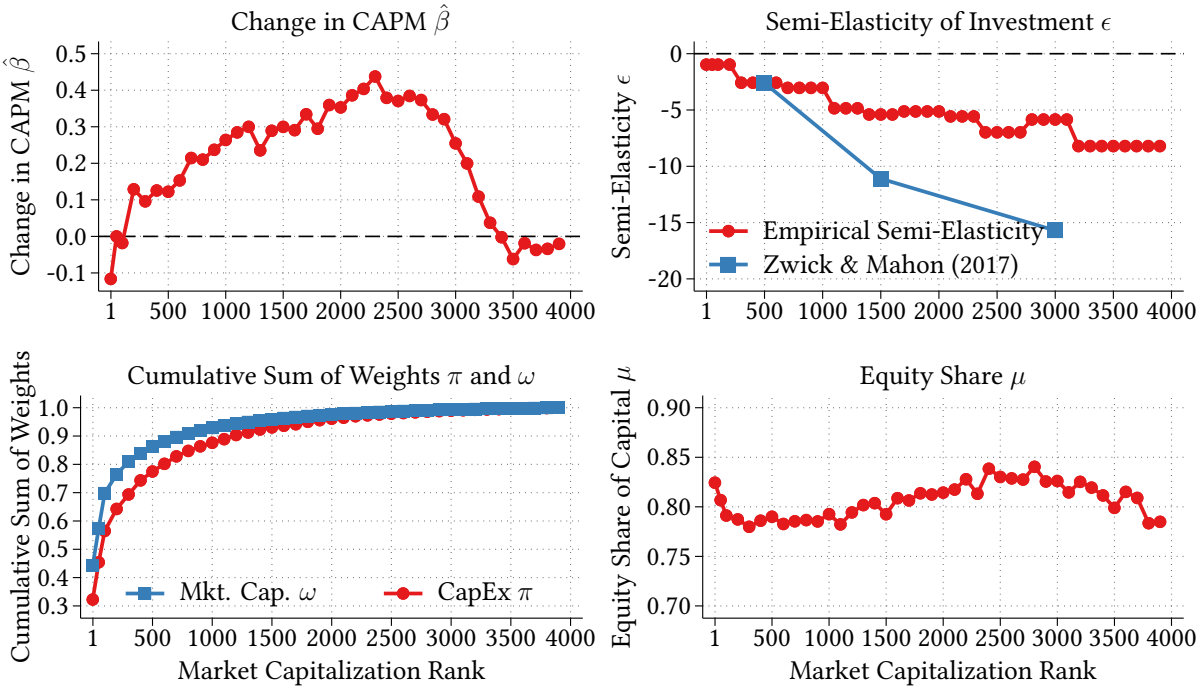
Notes: This figure shows local projection coefficients estimates for $100\times$ cumulative log-changes of outcome variables from regression of the form $\log(Y_{i,t+h}) - \log(Y_{i,t-1}) = \alpha^h_{j,t} + \gamma^h \widehat{\Delta \text{CAPM } \beta}_{i,t} + \phi^h \log(\text{Market Cap.})_{i,t} + \text{Banding Vars.}_{i,t} + \xi^h X_{i,t} + \varepsilon_{i,t+h}$, where the shock is scaled to a 0.25 increase in CAPM $\hat{\beta}$, which translate into a 100 bps rise in the CAPM-implied WACC ($0.25 \times 6\% \times \frac{2}{3}$) estimated with change in BMI as an instrumental variable. We restrict the estimation sample to stocks within 300 ranks around Russell index cutoffs. Dashed red lines represent 90% significance bands for the null of zero treatment effect, computed by inverting the F-statistic of joint significance around zero using Scheffé's method.

Figure B9: Difference-in-difference Event Study of MSCI ACWI Inclusion on CAPM $\hat{\beta}$



Notes: This figure shows dynamic treatment effects of MSCI ACWI benchmark inclusion on CAPM $\hat{\beta}$s to the local market index. Sample from 1999 to 2018, covering 23 developed and 24 emerging markets. The left panel uses a 252-day rolling β; the right panel uses a 60-month rolling β estimator. Specifications include year-by-industry-by-country fixed effects and log market capitalization, bid-ask spread, momentum, and log trading volume. Data from Jensen et al. (2023). Pointwise confidence intervals (95%) based on double-clustered standard errors.

Figure B10: Change in $\hat{\beta}$, Investment Semi-Elasticity, Aggregation Weights, and Equity Share



C Russell Index Methodology and Russell Identification

The Russell Index Reconstitution Process The Russell 1000, 2000, and 3000 indices are reconstituted annually in June based on a two-step process. First, index assignment is determined by ranking eligible securities by their total market capitalization on the designated "rank day" in May.[1] Before 2007, the assignment rule was simple: firms ranking in the top 1,000 formed the Russell 1000, while firms ranking from 1,001 to 3,000 formed the Russell 2000. This created a sharp cutoff at rank 1,000, where assignment is considered quasi-random.

In 2007, FTSE Russell introduced a "banding" policy around the 1,000th rank to reduce index turnover. Under this policy, a stock's assignment also depends on its index membership in the prior year, creating a band around the cutoff. Under the banding policy, a stock's assignment depends on its prior-year index membership:

- A stock from the previous year's Russell 2000 is assigned to the Russell 1000 if its market cap rank is between 1 and $1000 - c_1$.
- A stock from the previous year's Russell 1000 is assigned to the Russell 2000 if its market cap rank is between $1000 + c_2$ and 3,000,

otherwise the stock remains in its current index.

The band of stocks between ranks 1000 - c_1 and 1000 + c_2 constitutes a 5% band around the cumulative market cap of the largest 1000 stocks in the Russell 3000E universe. And then c_1 and c_2 are chosen such that the cumulative market cap of stocks ranked 1 through 1000 - c_1 is 95% of the cumulative market cap of the largest 1000 stocks, and the cumulative market cap of stocks ranked 1 through 1000 + c_2 is 105% of the cumulative market cap of the largest 1000 stocks. Note that even after the introduction of the banding policy, assignment to the Russell 1000 or 2000 is still based on a mechanical rule that changes each year with the distribution of firm sizes. We emphasize that there is no banding policy at the 3,000th rank cutoff (see, e.g., Section 6.10.3 on p. 19 of Russell (2025) US Equity Indexes Construction and Methodology).

Second, index weighting is based on each stock's float-adjusted market capitalization. Russell applies a proprietary, non-public float factor to determine the final weight. A larger float adjustment leads to a lower index weight, which corresponds to a lower BMI.

Empirical Strategy and Validation The official market capitalization ranks used for assignment are proprietary. We therefore construct a proxy for each firm's rank using public data,

[1]For most years, rank day is the last trading day in May. The full list of rank days from 1989 to 2019 is available in Appendix A of Ben-David et al. (2019).

Table C8: Predicting Russell 2000 Membership in June using Market Capitalization Rank Proxy

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: Russell 2000 Membership in June					
	Window width: 300 ranks			Window width: 150 ranks		
	1998-2018	1998-2006	2007-2018	1998-2018	1998-2006	2007-2018
$\mathbb{1}\{\text{Rank} > \text{cutoff in May}\}$	0.843***	0.924***	0.771***	0.830***	0.867***	0.751***
	(0.056)	(0.016)	(0.036)	(0.058)	(0.031)	(0.035)
Year Fixed Effect	✓	✓	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓	✓	✓
Banding Controls	✓		✓	✓		✓
Adj. R^2	0.91	0.95	0.94	0.88	0.92	0.92
Observations	15,032	4,456	10,130	8,721	2,183	6,268

Notes: This table reports estimates from regressions of the form:

$$\text{Russell 2000 Member in June}_{i,t} = \alpha_t + \mathbb{1}\{\text{Rank} > \text{cutoff in May}\}_{i,t} + \text{Baseline controls}_{i,t} + \text{Banding controls}_{i,t} + \varepsilon_{i,t}.$$

The dependent variable is an indicator for whether stock i is a member of the Russell 2000 index in June of year t. Baseline controls are the log of market capitalization and bid-ask spread measured in May of year t. Banding controls are indicator variables for having rank-date market cap in the band, an indicator for being in the Russell 2000 in May before reconstitution, and the interaction of these indicators. All regressions include year-fixed effects. We construct the market capitalization rank variable following Ben-David et al. (2019). Columns (1)-(3) use a window width of 300 ranks around the Russell 1000/2000 cutoff, while columns (4)-(6) use a window width of 150 ranks. We are grateful to Aditya Chaudhry for sharing his code and data which greatly helped us in constructing the market capitalization rank variable. * p<0.10, ** p<0.05, *** p<0.01.

following the methodology of Ben-David et al. (2019). To account for the post-2007 rule changes, our regression models include a set of banding controls as proposed by Appel et al. (2019). These controls include indicator variables for a firm's rank falling within the band, its membership in the Russell 2000 in the year prior, and the interaction between these two.

We validate our constructed rank proxy by confirming its ability to predict official Russell 2000 membership. As shown in Table C8, our proxy performs remarkably well. Our results are consistent with those of Pavlova & Sikorskaya (2023, Table 12), who conduct a similar analysis using the proprietary Russell ranks. Specifically: For the pre-banding period (1998–2006), our estimates are within one standard error of their findings. For the post-banding period (2007–2018), our estimates are lower but closely track theirs. Notably, even Pavlova & Sikorskaya (2023) report a predictive coefficient of only 0.85 in their post-banding regressions, underscoring that our proxy captures the vast majority of the explainable variation. Our results also align with similar validation exercises in Chaudhry (2025) and Ben-David et al. (2019).

To further validate our identification strategy, we replicate key results from Pavlova & Sikorskaya (2023). Specifically, we examine whether stocks with larger changes in BMI experience

Table C9: Replication of Pavlova & Sikorskaya's (2023) Price Pressure Result – BMI Change and Returns in April-July

	Total stock return in period				
Period	April	May	**June**	July	April-July
Panel A: Original Estimates of Pavlova & Sikorskaya (2023) from their Table A.19					
ΔBMI	0.151*	0.192***	**0.271****	0.037	0.677***
	(0.075)	(0.055)	**(0.102)**	(0.116)	(0.184)
Observations	14,547	14,547	14,547	14,547	14,547
Panel B: Replication of Estimates Using Ben-David et al. (2019) Ranking Variable Proxy					
ΔBMI	0.241**	0.186***	**0.267*****	0.035	0.787***
	(0.117)	(0.068)	**(0.102)**	(0.140)	(0.260)
Observations	13,232	13,233	13,232	13,157	13,266
p-value (Panel B = Panel A)	0.45	0.94	**0.97**	0.99	0.68

Notes: Panel A reproduces Pavlova & Sikorskaya (2023, Table A.19), which contains the main result for the month of June and provides additional evidence for anticipatory price pressures in months before the reconstitution. Panel B replicates these results in our sample using our constructed market capitalization rank variable based on Ben-David et al. (2019). The p-values test the null hypothesis that our coefficient estimates on ΔBMI in Panel B equal those reported by Pavlova & Sikorskaya (2023, Table A.19) in Panel A. Panel B reports estimates from regressions of the form:

$$\text{Return}_{i,t}^{\text{Month}} = \alpha_t + \gamma \Delta \text{BMI}_{i,t} + \text{Baseline controls}_{i,t} + \text{Banding controls}_{i,t} + \varepsilon_{i,t}.$$

Baseline controls are the log of market capitalization and bid-ask spread measured in May of year t. Banding controls are indicator variables for having rank-date market cap in the band, an indicator for being in the Russell 2000 in May before reconstitution, and the interaction of these indicators. All regressions include year fixed effects. We construct the market capitalization rank variable following Ben-David et al. (2019) and estimate the regressions using a window width of 300 ranks around the Russell 1000/2000 cutoff. Standard errors in parentheses are double-clustered by stock and year. * p<0.10, ** p<0.05, *** p<0.01.

differential returns around the Russell reconstitution period. Following their approach, we estimate regressions of the form:

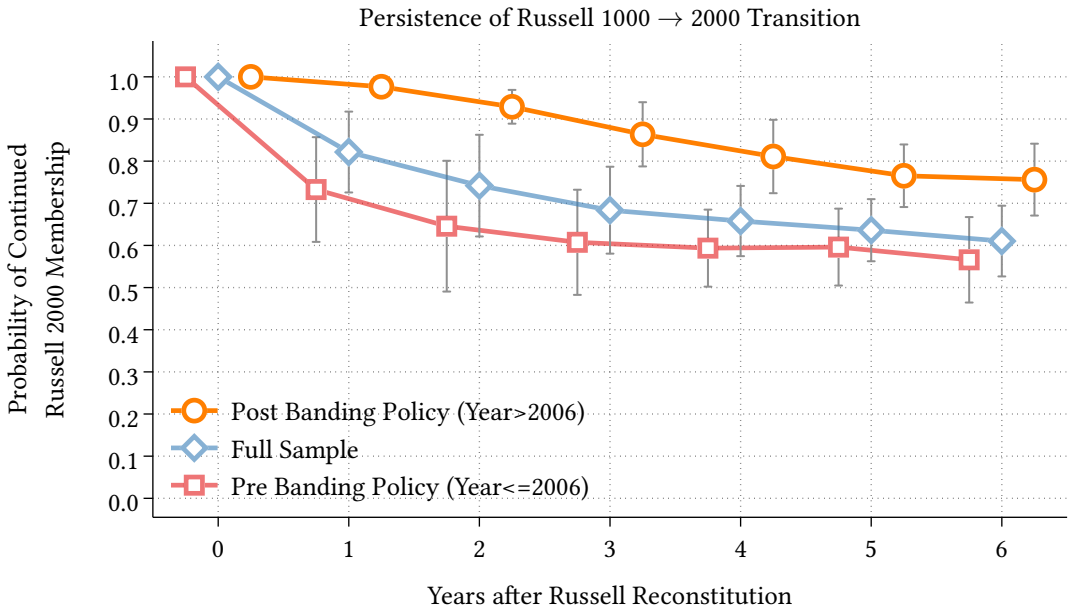
$$\text{Return}_{i,t}^{Month} = \alpha_t + \gamma \Delta BMI_{i,t} + \text{Baseline controls}_{i,t} + \text{Banding controls}_{i,t} + \varepsilon_{i,t}$$

where $\text{Return}_{i,t}^{\text{Month}}$ is the stock return for firm i in a given month, and $\Delta\text{BMI}_{i,t}$ is the change in benchmarking intensity.

Table C9 presents our replication results. Panel A reproduces the main findings from Pavlova & Sikorskaya (2023): stocks with larger increases in BMI experience significantly higher returns in May and June, with cumulative returns of 0.677% per 1 p.p. increase in ΔBMI over the April-July period. These results are consistent with price pressures from index fund rebalancing around the June reconstitution date.

Panel B shows our estimates using the market capitalization rank variable we constructed. To

Figure C11: Persistence of Transition from Russell 1000 to Russell 2000



Notes: This figure plots the probability of continued Russell 2000 membership in year $t + h$ after a transition from the Russell 1000 into the Russell 2000 in year t, estimated via Logit. Standard errors are clustered at the firm-level.

formally assess the accuracy of our replication, we test the null hypothesis that our coefficients equal theirs. The resulting p-values (all exceeding 0.45) indicate that we cannot reject equality between the two sets of estimates. This close replication validates our data construction methodology and confirms that our BMI measure accurately captures benchmark assignment dynamics.

These findings also align with the suggestive evidence in Pavlova & Sikorskaya (2023) (Appendix A.19) of anticipatory price pressures in months before reconstitution, demonstrating that at least some part of the total change in BMI is expected by markets prior to the official June reconstitution date.

Figure C11 shows that firms which transition from the Russell 1000 into the Russell 2000 have a high probability of staying in the Russell 2000 for several years. The conditional probabilities are estimated via Logit in our Compustat sample. The probability of staying in the Russell 2000 is higher after Russell introduced its banding policy in 2007. After 2007, a firm has a 76% of probability of staying in the Russell 2000 for more than 6 years, whereas before the introduction of the banding policy the probability was around 57%.

D Three period model with heterogeneous benchmarks

To illustrate our main mechanism, we develop a simple model of asset prices and investment in the presence of heterogeneous benchmarks. Our model introduces two new frictions into an otherwise standard model of benchmark-linked investing adapted from the literature (Brennan, 1993, Kashyap et al., 2021, Buffa et al., 2022, Pavlova & Sikorskaya, 2023). The first friction is uncertainty in the relative demand of benchmarked investors that proxies for fund flows. Exposure to benchmarked fund flows creates an additional source of comovement among benchmark constituent stocks that shifts the cross-section of CAPM βs. The second friction is a behavioral assumption about managers: we assume that managers do not observe the fundamental risk exposure of their firm and instead use the observed covariance of stock returns with the market to infer their cost of capital.

We proceed in two steps: First, we derive asset prices and the optimal investment policy with and without benchmarked investors. We illustrate that benchmarking distorts prices and expected returns relative to the CAPM and confirm the existence of a benchmark inclusion subsidy that lowers the costs of capital of firms included in a benchmark index. Next, we show that i) managers can infer the CAPM-optimal investment policy without knowledge of their fundamental risk exposure from the covariance of their stock returns with the market; ii) if assets are priced by benchmarked investors there is a two-factor structure in returns that increases the comovement of benchmark constituent stocks; iii) if managers erroneously implement the CAPM-optimal investment implied by the covariance of their stock returns with the market, they under-invest.

Setup There are three periods - denoted 0, 1, and 2 - and three types of agents: managers, investors and households. Each firm manager operates one of n fully equity-financed firms. Their objective function is to maximize firm value $\overline{x}_i S_{it}$ where S_{it} denotes the share price of the firm's equity and $\overline{x}_i$ denotes the number of outstanding shares normalized to $\overline{x} = \mathbf{1}$.

Investor demand The equity of each firm is priced by a unit mass of investors with absolute risk aversion γ. Their objective function is to maximize expected utility over next period wealth $U(W) = e^{-\gamma W}$ by investing in firms' (risky) equity and a risk-free bond that is in infinitely elastic supply with net interest rate normalized to 0. We follow the convention of the literature and denote the absolute returns of a portfolio $x_t = (x_t^1, \ldots, x_t^n)' \in \mathbb{R}^n$ by $R_{t+1}^x = x_t' R_{t+1}$, where $R_{t+1} = Y_{t+1} - S_t$ denotes the vector of absolute returns that is distributed with conditional covariance matrix Σ_t and $Y_{t+1} = S_{t+1} + D_{t+1}$ denotes the vector of total marketable cash flows.

Among investors, a fraction $\lambda_t^B \in [0, 1)$ invests funds on behalf of a principal. Their income

is contingent upon performance relative to a benchmark B represented by a benchmark portfolio vector $\mathbf{1}_B \in \{0,1\}^n$ with absolute returns $R^B_{t+1} = \mathbf{1}'_B(Y_{t+1} - S_t)$. Their compensation contract w is parameterized by constants $a \geq 0, c > 0$ and k.

$$w_t = aR^x_t + c(R^x_t - R^B_t) + k$$

The first and second term represent compensation for absolute performance R^x_t and performance relative to the benchmark portfolio R^B_t, respectively. The final term is a constant "base salary."[2]

The remaining $1 - \lambda^B_t$ investors are "fundamental" investors who invest directly and only care about absolute returns. The solution to their portfolio choice problem is standard:

$$x^D_t = \Sigma^{-1}_t \frac{\mathbb{E}_t\left[Y_{t+1}\right] - S_t}{\gamma} \tag{1}$$

The solution to the portfolio choice problem of a benchmarked investor is a combination of the mean-variance optimal portfolio x^D_t and the benchmark portfolio $\mathbf{1}_B$

$$x^B_t = \frac{1}{a+c}\Sigma^{-1}_t \frac{\mathbb{E}_t\left[Y_{t+1}\right] - S_t}{\gamma} + \frac{c}{a+c}\mathbf{1}_B \tag{2}$$

As first shown by Brennan (1993) benchmarking exposes investors to additional income risk that can be hedged by holding the benchmark portfolio. The more weight the compensation contract places on performance relative to the benchmark, the more weight a benchmarked investor places on the benchmark portfolio. In the limit, as $c \to \infty$, the portfolio of a benchmarked investor approaches the index itself (as in a passive fund evaluated on its tracking error).

Benchmarking Intensity We allow for heterogeneous benchmark portfolios across investors: There is a finite set of benchmark portfolios $\mathcal{B} = \{B_1, B_2, ...\}$, summarized by the matrix $\mathbf{1}_\mathcal{B} \in \{0,1\}^{n \times |\mathcal{B}|}$. The mass of benchmarked investors λ^B_t is partitioned into $|\mathcal{B}|$ disjoint masses $\lambda^\mathcal{B}_t = (\lambda^{B_1}_t, \lambda^{B_2}_t, ...) \in \mathbb{R}^{|\mathcal{B}|}_{\geq 0}$ with $\sum \lambda^{B_j}_t = \lambda^B_t$. This allows us to define the vector of benchmarking intensities across stocks as

$$BMI_t \equiv \sum_{B_j \in \mathcal{B}} \mathbf{1}_{B_j} \lambda^{B_j}_t = \mathbf{1}_\mathcal{B} \lambda^\mathcal{B}_t$$

[2] Kashyap et al. (2023) derive this specification as the solution to an agency problem. Studies of the mutual fund industry provide empirical support for this specification (Ma et al., 2019, Evans et al., 2024).

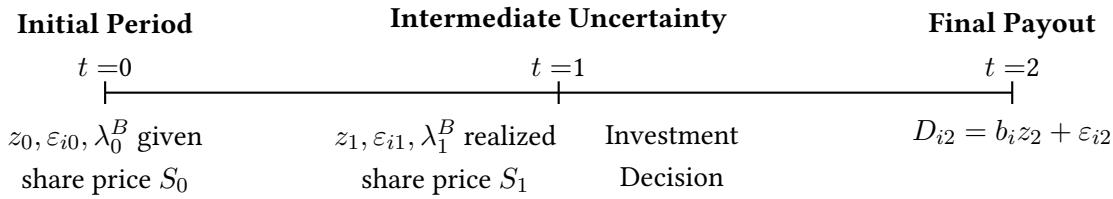


Figure D12: Model Timing

Timing The remaining setup is as follows: In period 2 all firms are liquidated with payout

$$D_{i2} = b_i z_2 + \varepsilon_{i2}$$

where $b \in \mathbb{R}^n_{\geq 0}$ is a vector of positive factor loadings on a common fundamental factor z_t, and ε_{it} is a firm specific idiosyncratic productivity process. Both the common factor and the idiosyncratic shock follow a square root process (as in Buffa et al., 2022, Kashyap et al., 2021),[3]

$$\begin{aligned} z_{t+1} &= z_t + \sqrt{z_t}\sigma_z u^z_{t+1}, \quad u^z_{t+1} \sim \mathcal{N}(0,1) \\ \varepsilon_{it+1} &= \varepsilon_{it} + \sqrt{\varepsilon_{it}}\sigma_e u^\varepsilon_{it+1}, \quad u^\varepsilon_{it+1} \sim \mathcal{N}(0,1) \end{aligned}$$

In period 0 the initial state of the economy $(z_0, \mathbf{e}_o, \lambda_0^B)$ is determined. Period 1 is divided into two subperiods: At the beginning of period 1 there is news about the state of the economy $(u_1^z, \mathbf{u}_1^\varepsilon)$ and a series of correlated shocks to the composition of investors $\lambda_0^\mathcal{B}$. Shocks to $\lambda^\mathcal{B}$ can be interpreted as net fund flows that induce correlated demand shocks – a positive shock to λ^B represents a flow of capital from active funds into benchmarked funds who invest the inflows according to their benchmark. The distribution of shocks to investor composition is governed by a correlation matrix $\Omega_\lambda \in [0,1]^{|\mathcal{B}|\times|\mathcal{B}|}$ and a scale parameter σ_λ.

$$\lambda_1^\mathcal{B} = \lambda_0^\mathcal{B} + \sigma_\lambda \mathbf{u}_1^\lambda, \quad \mathbf{u}_1^\lambda \sim \mathcal{N}(\mathbf{0}, \Omega_\lambda)$$

In response to the realization of shocks in period 1 managers are given the opportunity to invest in expanding the operations of their firm by a scalar μ such that their final payout is equal to $\widehat{D}_{i2}(\mu) = (1+\mu)D_{i2}$. Investment incurs a constant per-unit cost μC that managers finance by issuing δ additional shares of equity worth $\delta \widehat{S}_{i1}(\mu) = \mu C$, where $\widehat{S}_{i1}(\mu)$ denotes the

[3]The common caveat applies that the term in the square root cannot be negative (see Backus et al., 2001, for further discussion.). We assume that starting values $(z_0, \mathbf{e}_o, \lambda_0^B)$ and magnitude of shocks are such that the probability of either process dropping below zero is negligible.

value of the firm post-investment. The covariance matrix of payoffs post-investment is given by $\widehat{\Sigma}_1(\mu) = Var_1\left((1+\mu e_i)D_2\right)$ where e_i denotes the i-th unit vector.

Asset prices and optimal investment with benchmarking As shown by Kashyap et al. (2021) benchmarking distorts asset prices and investment policies so that the CAPM no longer holds. Imposing market clearing yields stocks prices

$$S_t = \mathbb{E}_t\left[Y_{t+1}\right] - \gamma\Lambda_t\Sigma_t\Big(\mathbf{1} - \frac{c}{a+c}BMI_t\Big) \tag{3}$$

where $\gamma\Lambda_t = \gamma\left[1-\lambda_t^B + \lambda_t^B/(a+c)\right]^{-1}$ modifies the market's effective coefficient of risk aversion. Equation (3) can be rearranged to construct an expression for expected returns

$$\mathbb{E}_t\left[R_{it+1}\right] = \beta_{it}^{CAPM}\mathbb{E}_t\left[\mu_{t+1}^M\right] - \beta_{it}^{\mathcal{B}}\mathbb{E}_t\left[\mu_{t+1}^{\mathcal{B}}\right] \tag{4}$$

where $\mathbb{E}_1\left[\mu_{t+1}^M\right] = \gamma\Lambda_t Var_t(\sum_j R_{jt+1})$ and $\beta_{it}^{CAPM} = \frac{Cov_t(R_{it+1},\sum_j R_{jt+1})}{Var_t(\sum_j R_{jt+1})}$ denote the market risk premium and CAPM β of firm i. Equations (3) and (4) cleanly illustrate the key frictions of our model: The inelastic demand of benchmarked investors raises the price of benchmark constituent stocks and lowers expected returns moving forward by introducing a second benchmarking subsidy factor in expected returns with factor loadings $\beta_{it}^{\mathcal{B}} = \frac{Cov_t(R_{it+1},\sum_j BMI_{jt}R_{jt+1})}{Var_t(\sum_j BMI_{jt}R_{jt+1})}$ and a negative risk premium $\mathbb{E}_t\left[\mu_{t+1}^{\mathcal{B}}\right] = \frac{c}{a+c}\gamma\Lambda_t Var_t(\sum_j BMI_{jt}R_{jt+1})$. This distorts the textbook single-factor CAPM relationship between expected returns and market risk exposure that otherwise holds if there is no benchmarking ($c=0$).

Proposition 1. *There is a benchmark inclusion subsidy that is increasing in benchmarking intensity and lowers the required return on investment for firms included in a benchmark index.*

As in standard investment models, the optimal investment policy is to invest if this increases firm value. This requires that the expected returns to investment are above a threshold $\underline{R}(BMI_{i1})$ that is decreasing in benchmarking intensity. In the limit, as $n\to\infty$, the optimal policy is to invest if the expected return per unit of investment $\mathbb{E}_1\left[D_{i2}\right] - C$ satisfies

$$\mathbb{E}_1\left[D_{i2}\right] - C > \beta_{i1}^{CAPM}\mathbb{E}_1\left[\mu_2^M\right]\left(1 - \frac{c}{a+c}BMI_{i1}\right) \tag{5}$$

Intuitively, the inelastic demand of benchmarked investors lowers Benchmarking gives managers of benchmark constituent firms an additional incentive to invest and lowers the required return on investment. The higher the benchmarking intensity of a firm, the larger the subsidy.

Asset prices and investment policy without benchmarking In the absence of benchmarking all assets are priced by fundamental investors and managers no longer need to take into account the inelastic demand of benchmarked investors. Without the benchmark inclusion subsidy the investment policy in (5) simplifies to a textbook CAPM investment policy: managers should invest if

$$\mathbb{E}_1\left[D_{i2}\right] - C > \tilde{\beta}_{i1}^{CAPM}\mathbb{E}_1\left[\tilde{\mu}_2^M\right] \tag{6}$$

where $\tilde{\beta}_{i1}^{CAPM}$ and $\mathbb{E}_2\left[\tilde{\mu}_2^M\right]$ denote the CAPM β of firm i and the market risk premium in the absence of benchmarked investors.

Assumption 1. *Managers do not observe their firm-specific factor loading b_i or the level of the common factor z_t, but they do observe aggregate growth $g_z \equiv z_1/z_0$, as well as their own idiosyncratic productivity $\varepsilon_{i,t}$, and assume that assets are priced by fundamental investors (the CAPM holds).*

This assumption represents the key behavioral friction of our model. As in practice, managers have limited information and cannot directly observe their fundamental risk exposure. Instead, they need to infer the required return on capital for their investment decisions, in the same way that practitioners use observable stock returns to infer their cost of equity.

Proposition 2. *If assets are priced by fundamental investors, managers can implement the CAPM-optimal investment policy in (6) without knowledge of their fundamental factor loading b_i or the* level *of the common factor z_t by observing the past covariance of stock returns with the market, $Cov_0\left(\tilde{R}_{i1}, \sum_j \tilde{R}_{j1}\right)$, growth in fundamentals $\frac{z_1}{z_0}$, and their idiosyncratic productivity $\varepsilon_{i0}, \varepsilon_{i1}$.*

In appendix D.1 we prove that there are terms[4] $A_{i1}(\frac{z_1}{z_0}, e_{i1}, e_{i0})$ and $\Gamma_1(\frac{z_1}{z_0})$ that allow to express the required return on investment in (6) as an affine function of the past covariance of returns with the market $Cov_0(\tilde{R}_{i1}, \sum_j \tilde{R}_{j1})$. Consequently, even a manager without knowledge of their own common factor loading b_i or the *level* of the common factor z_t can implement the investment policy in (6). They should invest if

$$\mathbb{E}_1\left[D_{i2}\right] - C > \gamma\Big[A_{i1} + \Gamma_1 Cov_0(\tilde{R}_{i1}, \sum_j \tilde{R}_{j1})\Big] \tag{7}$$

CAPM β and investment policy with benchmarked investors A first order approximation of prices with benchmarked investors S_1 in (3) around $\lambda_0^{\mathcal{B}}$ shows that up to $\mathcal{O}(\sigma_\lambda^2 (u_1^\lambda)^2)$ returns

[4]Specifically, $A_{i1} = \sigma_\varepsilon^2 \varepsilon_{i1} - \frac{(1-\gamma\sigma_\varepsilon^2)^2}{(1-\gamma\sigma_z^2)^2}\frac{z_1}{z_0}\sigma_\varepsilon^2\varepsilon_{i0}$ and $\Gamma_1 = \frac{1}{(1-\gamma\sigma_z^2)^2}\frac{z_1}{z_0}$

R_{i1} have a linear multi-factor structure.

$$R_{i1} \approx const. + \phi_i^z \Delta z_1 + \sum_{B_k \in \mathcal{B}} \phi^{\lambda}_{B_k(i)} \Delta \lambda_1^{B_k} + \nu_{i1} \tag{8}$$

The first factor with factor loadings ϕ_i^z reflects shocks to economic fundamentals. The second set of factors with loadings $\phi^{\lambda}_{B_k(i)}$ reflects shocks to the relative demand of benchmarked investors, such as fund flows. The loadings on the second factor consist of a systemic term that is independent of benchmark membership, and a benchmark-specific term that is increasing in benchmarking intensity.

Proposition 3. *Benchmarking shifts the distribution of CAPM βs and increases the variance of the market. An increase in benchmarking intensity raises (lowers) the CAPM β of benchmark constituents (non-constituents).*

Under the factor structure in (8) the CAPM β of a stock is equal to

$$\beta_{i0}^{CAPM} = \frac{Cov_0\left(\phi_i^z \Delta z_1 + \sum_k \phi^{\lambda}_{B_k(i)} \Delta \lambda_1^{B_k} + \nu_{i1}, \sum_j \phi_j^z \Delta z_1 + \sum_k \phi^{\lambda}_{B_k(j)} \Delta \lambda_1^{B_k} + \nu_{j1}\right)}{Var_0\left(\sum_j \phi_j^z \Delta z_1 + \sum_k \phi^{\lambda}_{B_k(j)} \Delta \lambda_1^{B_k} + \nu_{j1}\right)} \tag{9}$$

The introduction of a second factor that reflects the correlated demand shocks of benchmarked investors increases the variance of the market in the denominator for all stocks equally. However, the covariance term in the numerator varies with stocks benchmarking intensity. Benchmarking raises the covariance with the market for *all* stocks, but it has a larger effect for stocks with higher benchmarking intensities. On net, the CAPM β of stocks with the highest benchmarking intensity increases, while the CAPM β of stocks with lower benchmarking intensity decreases.[5]

Corollary 3.1. *All else equal, if managers pursue the CAPM-optimal investment policy in (7), compared to a setting without benchmarking i) all firms underinvest; ii) a firm inside the benchmark underinvests more than an otherwise identical firm outside the benchmark.*

If the CAPM holds managers can use the past covariance of stock returns with the market to implement the CAPM-optimal investment policy. The introduction of additional factors that amplify comovement, such as the demand of benchmarked investors, is perceived by managers as an increase on the required return on investment and induces them to under-invest relative to the CAPM. Because comovement increases more strongly for benchmark constituent stocks, this effect is particularly pronounced for stocks with high benchmarking intensity.

[5]By construction, the average CAPM β is constant and equal to 1.

D.1 Proofs

Combining the expressions for fundamental and benchmarked investor demand in (1) and (2) and imposing market clearing yields the expression for prices in (3). The aggregate demand of benchmarked and fundamental investors is equal to

$$\sum_{b\in\mathcal{B}} \lambda_t^B x_t^B + (1-\lambda_t^B)x_t^D = \Sigma^{-1}\left(1-\lambda_t^B+\frac{\lambda_t^B}{a+c}\right)\frac{\mathbb{E}_t\left[Y_{t+1}\right]-S_t}{\gamma} + \sum_{B_k\in\mathcal{B}}\frac{c}{a+c}\mathbf{1}_{B_k} \tag{10}$$

Imposing market clearing yields the expression for prices in (3).

Proposition 1. The total value of the firm post-investment is equal to

$$(1+\delta)\widehat{S}_{i1}(\mu) = (1+\mu)\mathbb{E}_1\left[D_{i2}\right] - \gamma\Lambda_1 e_i'\widehat{\Sigma}(\mu)\left(\mathbf{1}-\frac{c}{a+c}BMI_1\right)$$

which means that the change in firm value from investment is

$$\Delta\widehat{S}_{i1}(\mu) = \mu\left(\mathbb{E}_1\left[D_{i2}\right]-C\right) - e_i'\gamma\Lambda_1\left[\widehat{\Sigma}_1(\mu)-\Sigma_1\right]\left(\mathbf{1}-\frac{c}{a+c}BMI_{i1}\right)$$

Investment is profitable if a marginal unit of investment increases firm value, which requires that $\frac{\partial\Delta\widehat{S}_{i1}(0)}{\partial\mu} > 0$. This implicitly defines a lower bound on the required return on investment $\underline{R}(BMI_{i1})$ equal to

$$\mathbb{E}_1\left[D_{i2}\right]-C > \gamma\Lambda_1\underbrace{\left[(b'\mathbf{1}b_i+b_i^2)z_1\sigma_z^2+2\sigma_\varepsilon^2\varepsilon_{i1}\right]}_{=Cov_1\left(R_{i2},\sum_j R_{j2}\right)+Var_1(R_{i2})}\left(1-\frac{c}{a+c}BMI_{i1}\right) = \underline{R}(BMI_{i1})$$

To derive the limit result in (5) we consider a sequence of economies indexed by the number of firms n and investors' coefficient of absolute risk aversion $\gamma(n)=\frac{\gamma}{n}$ that scales with the size of the economy.[6] Firms' factor loadings b_i are drawn from a distribution $F(b)$ with positive support, finite variance and mean $\mu(b)$. As $n\to\infty$ the threshold on the right hand side converges to

$$\lim_{n\to\infty}\underline{R}(BMI_{i1}) = \beta_{i1}^{CAPM}\mathbb{E}_1\left[\mu_2^M\right]\left(1-\frac{c}{a+c}BMI_{i1}\right)$$

[6]This ensures that if we were to duplicate an economy with n firms the price of each firm is unaffected.

where

$$\beta_{i1}^{CAPM} = \lim_{n\to\infty} \frac{\frac{1}{n}Cov_1(R_{i2}, \sum_j R_{j2})}{\frac{1}{n}Var_1(\sum_j R_{j2})} = \frac{b_i \mu(b)\sigma_z^2 z_1}{\mu(b)^2\sigma_z^2 z_1}$$
$$\mathbb{E}_1\left[\mu_2^M\right] = \lim_{n\to\infty} \gamma(n)\Lambda_1 Var_1(\sum_j R_{j2}) = \gamma\Lambda_1\mu(b)^2\sigma_z^2 z_1$$

Proposition 2. If there is no benchmarking assets are priced by fundamental investors with mean-variance preferences and prices simplify to a standard textbook solution.

$$\tilde{S}_t = \mathbb{E}_t\left[\tilde{Y}_{t+1}\right] - \gamma Var_t\left(\tilde{Y}_{t+1}\right)\mathbf{1} \tag{11}$$

Rearranging (11) yields the textbook expression for the CAPM (in absolute returns):

$$\mathbb{E}_t\left[R_{it+1}\right] = \underbrace{\frac{Cov_t(\tilde{Y}_{it+1} - \tilde{S}_{it}, \sum_j \tilde{Y}_{jt+1} - \tilde{S}_{jt})}{Var_t\left(\sum_j \tilde{Y}_{jt+1} - \tilde{S}_{jt}\right)}}_{=\tilde{\beta}_{it}^{CAPM}} \underbrace{\gamma Var_t\left(\sum_j \tilde{Y}_{jt+1} - \tilde{S}_{jt}\right)}_{=\mathbb{E}_t\left[\tilde{\mu}_{t+1}^M\right]}$$

The optimal investment policy without benchmarking follows directly from (5) and setting $\lambda^B = 0$. In order to implement the investment policy in (6) it is sufficient for managers to have knowledge of $Cov_1(\tilde{R}_{i2}, \sum_j \tilde{R}_{j2}) = \tilde{\beta}_{i1}^{CAPM}\mathbb{E}_1\left[\tilde{\mu}_2\right]$ as opposed to both $\tilde{\beta}_{i1}^{CAPM}$ and $\mathbb{E}_1\left[\tilde{\mu}_2\right]$. In order to express $Cov_1(\tilde{R}_{i2}, \sum_j \tilde{R}_{j2})$ as an affine function of $Cov_0(\tilde{R}_{i1}, \sum_j \tilde{R}_{j1})$ note that if assets are priced by fundamental investors time 1 and time 0 prices are

$$S_1 = bz_1 + \varepsilon_1 - \gamma\Sigma_1\mathbf{1}$$
$$S_0 = b(1-\gamma\sigma_z^2)^2 z_0 + \varepsilon_0(1-\gamma\sigma_\varepsilon^2)^2 - \gamma\Sigma_0\mathbf{1}$$

where $\Sigma_1 = bb'\sigma_z^2 z_1 + \sigma_\varepsilon^2 Diag(\varepsilon_1)$ and $\Sigma_0 = bb'\left(1-\gamma\sigma_z^2\right)^2\sigma_z^2 z_0 + \left(1-\gamma\sigma_\varepsilon^2\right)^2\sigma_\varepsilon^2 Diag(\varepsilon_0)$. The conditional covariance of returns with the market in each period is equal to

$$Cov_1(D_{i2} - S_{i1}, \sum D_{j2} - S_{j1}) = \sigma_z^2 b'\mathbf{1}b_i z_1 + \sigma_\varepsilon^2\varepsilon_{i1}$$
$$Cov_0(S_{i1} - S_{i0}, \sum S_{j1} - S_{j0}) = \left[1-\gamma\sigma_z^2\right]^2\sigma_z^2 b'\mathbf{1}b_i z_0 + \left[1-\gamma\sigma_\varepsilon^2\right]^2\sigma_\varepsilon^2\varepsilon_{i0}$$

Rearranging and pre-multiplying the second line by $\frac{z_1}{z_0}$ illustrates that the time 1 covariance of future returns can be expressed in terms of the past covariance of stock returns with the market, $Cov_0\left(S_{i1} - S_{i0}, \sum_j S_{j1} - S_{j0}\right)$, growth in the common factor $\frac{z_1}{z_0}$, and idiosyncratic productivity

$\varepsilon_{i0}, \varepsilon_{i1}$:

$$Cov_1(D_{i2} - S_{i1}, \sum_j D_{j2} - S_{j1}) = A_{i1} + \Gamma_1 Cov_0(S_{i1}, \sum_j S_{j1}) \tag{12}$$

where

$$A_{i1} = \sigma_\varepsilon^2 \varepsilon_{i1} - \frac{(1-\gamma\sigma_\varepsilon^2)^2}{(1-\gamma\sigma_z^2)^2}\frac{z_1}{z_0}\sigma_\varepsilon^2\varepsilon_{i0}$$
$$\Gamma_1 = \frac{1}{(1-\gamma\sigma_z^2)^2}\frac{z_1}{z_0}.$$

Proposition 3. Given the expression for period 1 prices in (3) we can express S_{i1} as an affine function of z_1 and ε_{i1}

$$S_{i1} = \Theta_z(\lambda_1^{\mathcal{B}}) b_i z_1 + \Theta_\varepsilon(\lambda_1^{\mathcal{B}})\varepsilon_{i1}$$

with coefficients $\Theta_z(\lambda_1^{\mathcal{B}})$, $\Theta_{\varepsilon i}(\lambda_1^{\mathcal{B}})$ equal to

$$\Theta_z(\lambda_1^{\mathcal{B}}) = 1 - \gamma\Lambda(\lambda_1^{\mathcal{B}})\sigma_z^2\left(1 - \frac{c}{a+c}b' BMI_1(\lambda_1^{\mathcal{B}})\right)$$
$$\Theta_{\varepsilon i}(\lambda_1^{\mathcal{B}}) = 1 - \gamma\Lambda(\lambda_1^{\mathcal{B}})\sigma_\varepsilon^2\left(1 - \frac{c}{a+c}BMI_{1i}(\lambda_1^{\mathcal{B}})\right)$$

A first order Taylor approximation of S_{i1} around λ_0^B shows that up to order $\mathcal{O}(\sigma_\lambda^2(u_1^\lambda)^2)$

$$S_{i1} \approx [\Theta_z(\lambda_0^{\mathcal{B}}) + (\Delta\lambda_1^{\mathcal{B}})^T\nabla_\lambda^z] b_i(z_0 + \Delta z_1) + [\Theta_{\varepsilon i}(\lambda_0^{\mathcal{B}}) + (\Delta\lambda_1^{\mathcal{B}})^T\nabla_\lambda^{\varepsilon i}](\varepsilon_{i0} + \Delta\varepsilon_{i1})$$
$$\text{with } \nabla_\lambda^z(\lambda_0^{B_k}) = \gamma\Lambda_0\sigma_z^2\left[\frac{c}{a+c}b'\mathbf{1}_{B_k} - \Lambda_0\left(1 - b'\frac{c}{a+c}BMI_0\right)\left(1 - \frac{1}{a+c}\right)\right]$$
$$\nabla_\lambda^{\varepsilon i}(\lambda_0^{B_k}) = \gamma\Lambda_0\sigma_\varepsilon^2\left[\frac{c}{a+c}\mathbf{1}_{B_k}(i) - \Lambda_0\left(1 - \frac{c}{a+c}BMI_{0i}\right)\left(1 - \frac{1}{a+c}\right)\right]$$

which means that for small values of σ_λ^2 returns $S_{i1} - S_{i0}$ have a linear multi-factor structure in fundamental shocks Δz_1 and fund flows $\Delta\lambda_1^{B_k}$

$$S_{i1} - S_{i0} \approx const. + \phi_i^z \Delta z_1 + \sum_k [\phi_{k,S(i)}^\lambda + \phi_{k,B(i)}^\lambda]\Delta\lambda_1^{B_k} + \nu_{i1} \tag{13}$$

with factor loadings ϕ_i^z and $\sum_k[\phi^\lambda_{k,S(i)} + \phi^\lambda_{k,B(i)}]$.

$$\begin{aligned}
\phi_i^z &= b_i\left(1-\gamma\Lambda(\lambda_0^{\mathcal{B}})\sigma_z^2\left(1-\frac{c}{a+c}b'BMI_0\right)\right)\\
\phi^\lambda_{k,S(i)} &= \gamma\Lambda_0\left\{\frac{c}{a+c}\left[b'\mathbf{1}_{B_k}+\Lambda_0\left(1-\frac{c}{a+c}b'BMI_0\right)\right]b_i z_0\sigma_z^2-\Lambda_0\left(1-\frac{c}{a+c}\right)\left(b_i z_0\sigma_z^2+\varepsilon_{i0}\sigma_\varepsilon^2\right)\right\}\\
\phi^\lambda_{k,B(i)} &= \gamma\Lambda_0\left\{\frac{c}{a+c}\left[b'\mathbf{1}_{B_k}(i)+\Lambda_0\left(1-\frac{c}{a+c}\right)BMI_{0i}\right]\right\}\varepsilon_{i0}\sigma_\varepsilon^2
\end{aligned}$$

Benchmarking not only introduces a second factor but also changes the loadings on the fundamental factor. Plugging these expressions into the CAPM β that an econometrician would estimate yields

$$\begin{aligned}
\beta_{i0}^{CAPM} &= \frac{Cov_0\left(\phi_i^z\Delta z_1+\sum_k[\phi^\lambda_{k,S(i)}+\phi^\lambda_{B_k(i)}]\Delta\lambda_1^{B_k}+\nu_{i1},\sum_j\phi_j^z\Delta z_1+\sum_k[\phi^\lambda_{k,S(j)}+\phi^\lambda_{B_k(j)}]\Delta\lambda_1^{B_k}+\nu_{j1}\right)}{Var_0\left(\sum_j\phi_j^z\Delta z_1+\sum_k[\phi^\lambda_{k,S(j)}+\phi^\lambda_{B_k(j)}]\Delta\lambda_1^{B_k}+\nu_{j1}\right)}\\
&= \frac{\phi_i^z(\mathbf{1}'\phi^z)\sigma_z^2+[\phi^\lambda_{S(i)}+\phi^\lambda_{B(i)}]'\Omega_\lambda[\phi^\lambda_S+\phi^\lambda_B]\sigma_\lambda^2+\sigma^2_{\nu,i}}{(\mathbf{1}'\phi^z)^2\sigma_z^2+[\phi^\lambda_S+\phi^\lambda_B]'\Omega_\lambda[\phi^\lambda_S+\phi^\lambda_B]\sigma_\lambda^2+\sum_j\sigma^2_{\nu,j}}
\end{aligned}$$

Compared to a setting without benchmarked investors we observe that

1. The covariance of all stocks with the market increases
2. The covariance of stocks inside a benchmark index increases additionally by $\beta_{ik}^{\lambda_k,i}$
3. The variance of the market increases
4. A shift in initial exposure to investor composition shocks $\Delta\lambda_1^B$, for example due to adding an additional index or an increase in c shifts the initial cross-section of β_{i0}^{CAPM}.

E Simulated CAPM βs in a Two-Factor Model

This appendix uses a simulation to show that the emergence of a passive flow factor can explain the evolution of CAPM $\hat{\beta}$s between 1998 and 2018. We specify a parsimonious two-factor model where stock returns depend on: (1) a fundamental factor derived from macro-financial variables, with loadings fixed to the pre-benchmarking era (May 1990), and (2) a flow factor constructed from passive fund flows, with exposures proportional to stocks' benchmarking intensity.

The simulation yields three key findings: (i) The model successfully replicates both the cross-sectional distribution and time-series evolution of empirical CAPM $\hat{\beta}$s. (ii) Calibrating the model with actual passive fund flows accurately reproduces the observed conditional covariance structure and time series of $\hat{\beta}$s. (iii) In contrast, a model using active fund flows fails to match these empirical patterns, especially after 2003.

Model Suppose excess returns on stock i obey the following factor structure

$$R_{i,t+1} - R_t^f = a_{i,t} + b_{i,t}\lambda_{t+1} + u_{i,t+1} \tag{14}$$

where $\lambda_{t+1} = (z_{t+1}\ f_{t+1})'$ denotes "fundamental" and flow factors, and $(b_{it}^1\ b_{it}^2)$ denotes the loadings on the factors where b_{it}^2 is proportional to the benchmarking intensity of stock i. The covariance structure of the factors Σ_λ may contain positive off-diagonal elements, the covariance structure of idiosyncratic shocks Σ_u does not.

An econometrician estimating the CAPM β of a stock as $\hat{\beta} = \frac{Cov_t(R_{i,t+1}, R_{m,t+1})}{Var_t(R_{m,t+1})}$ where $R_{m,t+1} = \sum_j w_{j,t} R_{j,t+1}$ denotes the market capitalization weighted average return on the universe of securities j. Plugging (14) into the CAPM β formula, one finds that

$$\begin{aligned}\hat{\beta}_{it} &= \frac{Cov_t\left(b_{i,t}\lambda_{t+1} + u_{i,t+1}, \sum_j w_{jt}\left(b_{j,t}\lambda_{t+1} + u_{j,t+1}\right)\right)}{Var_t\left(\sum_j w_{jt}(b_{j,t}'\lambda_{t+1} + u_{j,t+1})\right)} \\ &= \frac{w_t' b_t \Sigma_\lambda b_t' e_i + w_{i,t}\sigma_{u,i}^2}{w_t' b_t \Sigma_\lambda b_t' w_t + w_t' \Sigma_u w_t}\end{aligned} \tag{15}$$

where e_i denotes the i-th unit vector.

To replicate Figure 1 in our simulation, we simulate Eq. (15) using the conditional means of CAPM $\hat{\beta}s$ across market capitalization ranks rather than individual stocks $\hat{\beta}_{it}$. We model conditional means of CAPM $\hat{\beta}$ and BMI as flexible fifth-order polynomial functions of market capital-

ization ranks.[7] We then compare the simulated conditional means to the empirical conditional means of CAPM $\hat{\beta}$s across market capitalization ranks. To evaluate model fit, we compute root mean square error (RMSE) and Spearman rank correlations in each cross-section. Finally, we examine various model calibrations to assess how the flow factor and covariance structure between factors affect the simulation outcomes.

Baseline Calibration To simulate Eq. (15), we need estimates of the factor loadings b_t, factor covariance matrix Σ_λ, idiosyncratic variances Σ_u, and weights w_{it}.

We start by calibrating b_t. We fix the fundamental factor exposures to $b^1_{it} = \text{CAPM } \hat{\beta}_{i1990m5} \; \forall \; t$. This ensures that time-variation in our simulated CAPM $\hat{\beta}$s is driven by exposure to the flow factor. We specify the flow factor loadings as proportional to the stock's benchmarking intensity : $b^2_{it} \propto \text{BMI}_{it}$. This reflects our hypothesis that stocks with higher benchmarking intensity experience greater exposure to benchmark-driven capital flows. The proportionality constant is difficult to precisely determine empirically. However, Table 1 provides evidence that net flows into passive funds predict changes in CAPM $\hat{\beta}$s. We thus set the proportionality constant to 0.3, matching the coefficient of the BMI-passive flow interaction term from Column 5 of Table 1.

We next calibrate the factor covariance matrix, Σ_λ. To do so, we first need to specify what the fundamental factor and flow factor are. We proxy the fundamental factor using the first principal component (PC) derived from various macro-financial variables: log changes in industrial production (Cochrane, 1991), the 3-month Treasury bill rate (Bernanke & Kuttner, 2005), unemployment rate (Kilic & Wachter, 2018), WTI oil price (Kilian & Park, 2009), the University of Michigan consumer sentiment index (Baker & Wurgler, 2006), and the consumer price index (Campbell & Ammer, 1993). The first PC explains 39.2% of the variation in these variables between 1990 and 2024. We scale the PC by 10^{-1} to align its scale to the $\hat{\beta}$s from May 1990.[8]

We specify the flow factor as net flows into passive mutual funds and ETFs, scaled by their total net assets. We compare the net flows into passive funds with net flows into active funds to determine whether flows in general or passive flows in particular drive changes in CAPM $\hat{\beta}$s. We estimate each component of the factor covariance matrix using 60-month rolling windows,

$$\widehat{\Sigma}_{\lambda,t} = \begin{pmatrix} \widehat{\sigma}^2_{z,t} & \widehat{\rho}_{zf,t}\widehat{\sigma}_{z,t}\widehat{\sigma}_{f,t} \\ \widehat{\rho}_{zf,t}\widehat{\sigma}_{z,t}\widehat{\sigma}_{f,t} & \widehat{\sigma}^2_{f,t} \end{pmatrix},$$

and set $\sigma_{u,it} = 0.07 \; \forall \; i, t$.

[7] Specifically, we the estimate conditional means each month as $y_i = \gamma_0 + \sum_{j=1}^{5} \gamma_j (\text{ME rank}_i)^j + \varepsilon$.

[8] Using changes in industrial production or the excess return on the market itself as the factor yields similar results.

Table E10: Model Evaluation – RMSE and Rank Correlation

	RMSE		Rank Correlation		Monthly
Model Calibration	Passive Flows	Active Flows	Passive Flows	Active Flows	Obs.
Baseline	**0.162**	**0.294**	**0.49**	**-0.02**	**249**
Full Sample $\widehat{\Sigma}_\lambda$	0.161	0.273	0.48	0.05	249
Fixed Weights ($w_{it} = w_{i1990m5}\ \forall i$)	0.159	0.286	0.49	-0.02	249
Flow Factor Off ($\sigma_f = 0$)	0.531	0.531	-0.07	-0.07	249

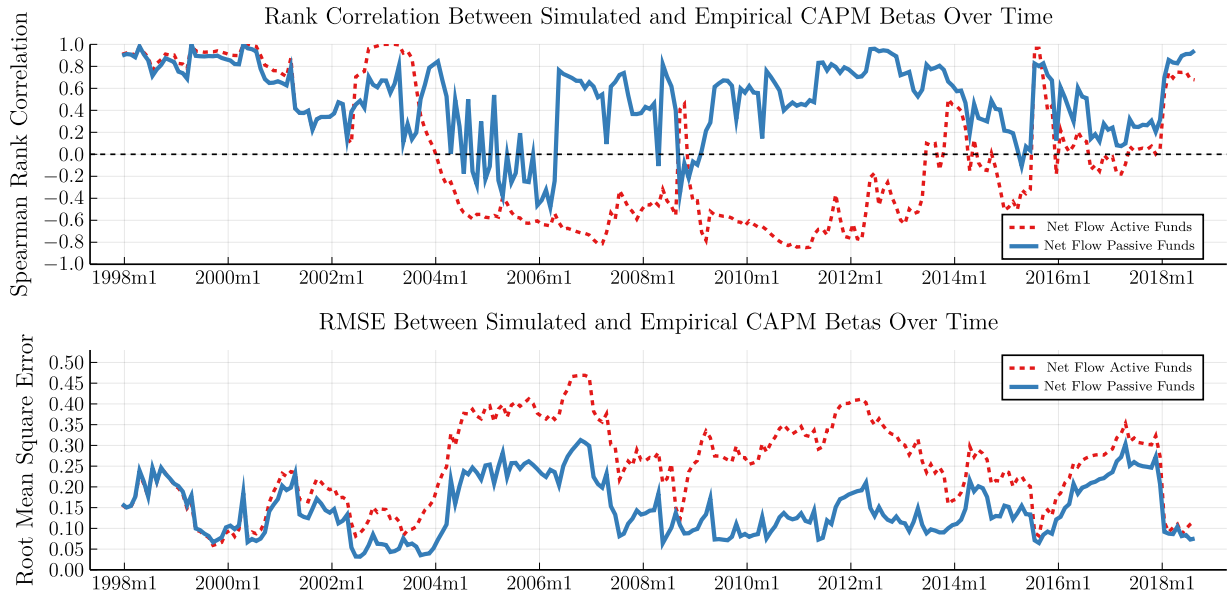
Notes: This table reports the average root mean square error (RMSE) and Spearman rank correlation between simulated and empirical CAPM $\hat{\beta}$s from January 1998 to September 2018. Each month, we simulate conditional means of CAPM $\hat{\beta}$ across market capitalization ranks and compare them with empirical conditional means. We report results for calibrations using active and passive net flows under different model configurations.

Results Our simulation yields three key insights: (i) Introducing a flow factor, whose loadings are proportional to benchmarking intensity, explains both the cross-sectional distribution and the temporal evolution of CAPM $\hat{\beta}$s from 1998 to 2018. (ii) Calibrating the flow factor using passive net flows allows us to match the observed cross-section and time-series of CAPM $\hat{\beta}$s. (iii) In contrast, calibration using active fund flows fails to reproduce these observed conditional moments. These findings support our hypothesis that passive flows are a key driver behind the observed increases in CAPM $\hat{\beta}$s.

Table E10 reports the average RMSE and Spearman rank correlations between simulated and empirical CAPM $\hat{\beta}$s for each month from January 1998 to September 2018. Comparisons across model calibrations highlight that active fund flows yield RMSE nearly twice as large as those from passive flows. The baseline calibration has an average RMSE of 0.16 (passive) versus 0.29 (active). Additionally, simulated cross-sectional distributions of CAPM $\hat{\beta}$s correlate strongly with actual distributions when calibrated to passive flows (avg. correlation of 0.49), whereas correlations using active flows are close to zero. Comparing different model specifications, results for passive flows remain robust. Eliminating the flow factor ($\sigma_f = 0$) worsens model performance, increasing RMSE dramatically and implies negative correlations between simulated and observed $\hat{\beta}$s. Figure E13 shows the time-series evolution of rank correlations and RMSE between simulated and empirical CAPM $\hat{\beta}$s from 1998 to 2018. The passive-flow calibration consistently outperforms the active-flow calibration, particularly after 2007. This aligns with Table 1, which indicates a stronger correlation between passive flows, benchmarking intensity, and $\hat{\beta}$s post-2010.

Figure E14 illustrates our two-factor model's ability to replicate CAPM $\hat{\beta}$ evolution across market capitalization ranks from 2000 to 2018. The solid blue line represents simulated conditional means, while the dashed red line shows empirical conditional means of CAPM $\hat{\beta}$s. Despite its simplicity, the model successfully captures key empirical patterns in the data. It is able to repli-

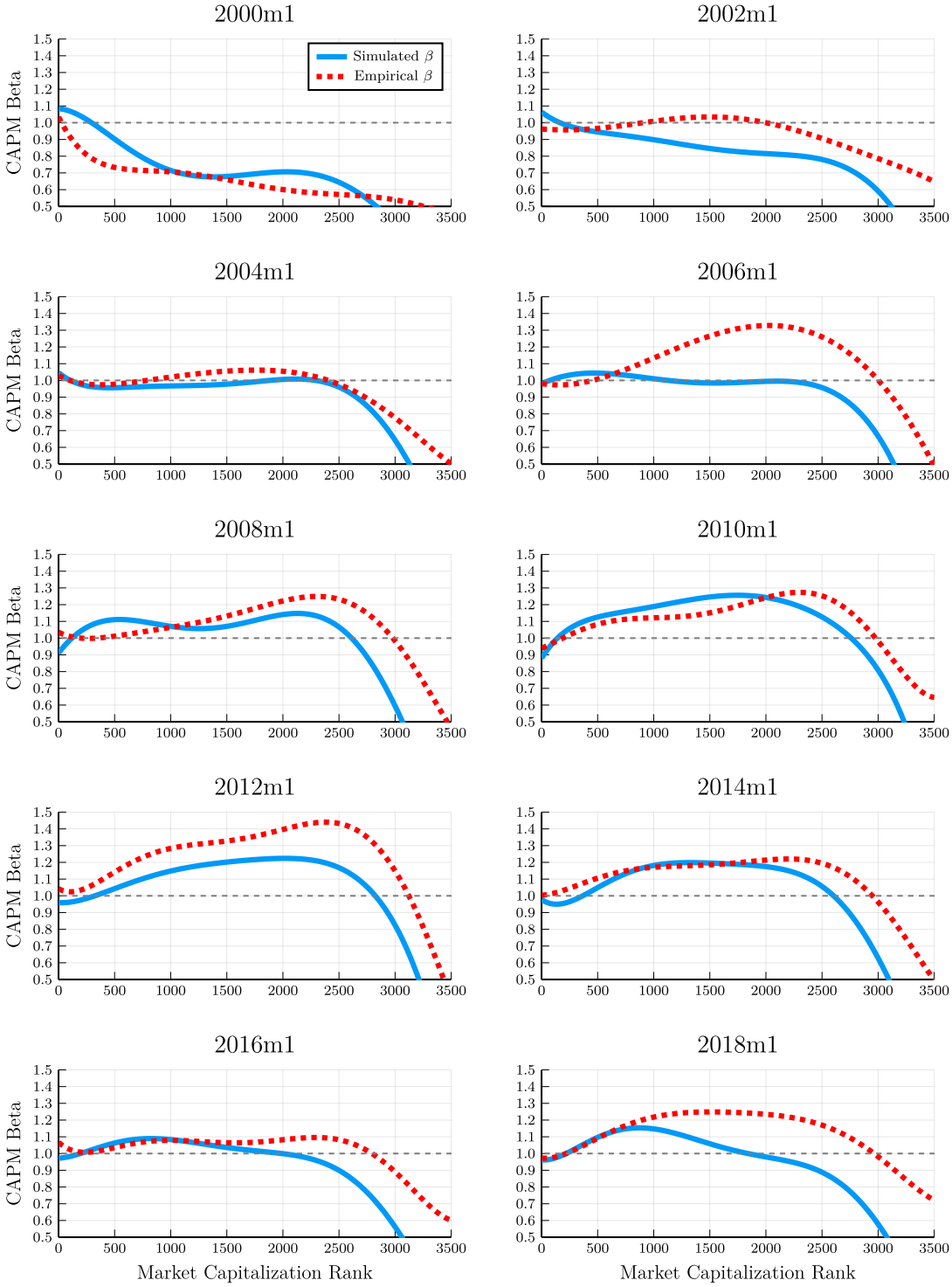
Figure E13: Time Series of Correlation and RMSE Between Simulated and Empirical CAPM $\hat{\beta}$s



Notes: This figure plots Spearman rank correlation and root mean square error between the empirical and simulated conditional means of CAPM $\hat{\beta}s$ from 2000 to 2018. Solid blue lines are report results from a calibration using net flows into *passive* mutual funds and ETFs. Dashed red lines are report results from a calibration using net flows into *active* mutual funds.

cate the cross-sectional distribution well. Moreover, the model is able to replicate the time-series increase in CAPM $\hat{\beta}$s across market capitalization ranks from 2000 to 2018.

Figure E14: Simulated and Empirical CAPM $\hat{\beta}$s Across Market Capitalization Ranks



Notes: This figure plots the empirical and simulated conditional means of CAPM $\hat{\beta}s$ from 2000 to 2018. We calibrate the flow factor to the net flows into passive mutual funds and ETFs. Dashed red line is the conditional mean of the empirical distribution across market capitalization ranks. Solid blue line is the conditional mean of the simulated distribution across market capitalization ranks.

F Qualitative Evidence from Earnings Call Transcripts

This appendix presents qualitative evidence from earnings calls on how corporate executives discuss Russell benchmark inclusion and their use of the CAPM to estimate the cost of equity.

F.1 Executives Commenting on Russell Benchmark Inclusion

To understand how corporate executives frame the importance of benchmark inclusion, we analyze earnings call transcripts from Capital IQ (via WRDS) between 2008 and 2024. Our search for terms related to Russell reconstitution yields 165 unique transcripts.[9] After excluding one transcript from the London Stock Exchange Group (owner of the Russell brand) and two from BlackRock (discussing index products), our sample consists of 163 transcripts. We classify these transcripts into four mutually exclusive categories based on the context of the mention. To ensure a conservative classification, we assign transcripts sequentially based on the following hierarchy:

1. **Equity Offerings (*5 transcripts*):** Discussion of an at-the-market, seasoned equity, follow-on, shelf takedown, or other equity offering associated with inclusion.
2. **Liquidity (*26 transcripts*):** Mention of increased stock liquidity or trading volume.
3. **Visibility (*34 transcripts*):** Reference to increased visibility or a broader investor base.
4. **Passing Mention (*98 transcripts*):** A brief statement of inclusion with no further context.

This analysis shows that executives rarely link Russell benchmark inclusion to capital-raising activities. Only 3% of transcripts mention equity offerings associated with the reconstitution. In contrast, the most common context is a passing mention (60% of cases), where the managers note inclusion as a recent corporate achievement without further elaboration. For example,

> "Top line data from our PEDFIC-1 Phase III trial is expected at the end of 2019 or early 2020. **And finally, we were selected for inclusion in the Russell 2000 Index Russell as part of the annual reconstitution of the Russell stock indexes.** And with that, let me turn the call back over to Ron for some concluding remarks."
>
> — CFO, Albireo Pharma Inc. (Q2 2018)

> "The first half of this year has been extremely productive, and we are driving continued progress, sustainable and profitable growth as a leader and the disruptor of the B2B e-commerce technology solution. **We were honored to be added to the Russell 2000 Index through their recent reconstitution**. Now I will turn the call over to Iman to provide more color and – on our operational highlights."
>
> — CEO, GigaCloud Technology Inc. (Q2 2024)

[9] We query: "russell+index+inclusion+(1000|2000|3000)" and "russell+index+reconstitution+(1000|2000|3000)."

In 26 instances, executives connect the benchmark inclusion to expected or realized improvements in stock liquidity and trading volume. For example,

> "All this happened while we were distracted by hurricanes, wildfires and mudslides. In addition, **we were relisted on the Russell 2000 Index, increased our average daily share volume by over 200%** and outperformed both our peers and the S&P 500. For the full year 2017, our revenue was up 39% [...]"
>
> —CEO, Sterling Construction Company (Q4 2017)

> "Before we move into our financial results, I want to quickly update you on some exciting corporate developments. First, we were notified in the end of June that **we were added to the Russell 3000 and Russell Global Indexes as part of Russell Investments' annual reconstitution**. We always prioritize building shareholder value and **we hope that this inclusion may improve general liquidity in our stock.** We also announced last week some changes to our Board of Directors [...]"
>
> — President, BioSpecifics Technologies Corp (Q2 2014)

A third prominent theme, found in 34 transcripts, is the belief that benchmark inclusion serves as a catalyst for increased corporate visibility. For example,

> "As you may have seen in June, we have been added to the Russell 2000 index at the conclusion of the indexes' annual reconstitution. We are proud to have come this far as a public company and expect this new index addition to provide **increased awareness to the broader investment community going forward."**
>
> —CEO, StarTek, Inc. (Q2 2019)

> "Working with our leadership team Carol will implement our company's strategic initiatives and increase operational execution across all our business channels. And **we are very pleased to rejoin the Russell 2000 index following a reconstitution on June 24. We believe this addition will bring additional visibility to our company and our strategic initiatives.** Now I'd like to turn the call over to David C., to provide more details on our financial performance for the second quarter."
>
> — CEO, Hickory Tech Corp. (Q2 2011)

F.2 Executives Referencing The CAPM to Calculate Cost of Equity

We perform a similar analysis and search earnings call transcripts from Capital IQ for mentions of "capital asset pricing model" or "CAPM." Corporate executives frequently reference the CAPM in response to analysts' questions about their firm's cost of equity and capital budgeting decisions:

> "Why do we think that a 10% return is good? Well, you have to — whether we're creating shareholder value really goes to what's our cost of capital. [...] this is really how we view our weighted average cost of capital. Most of you will bring back visions of business school. This is the **capital asset pricing model**, right? Our cost of equity, about 10.7%."
>
> — CFO, LKQ Corp. (Q1 2016)

> "If you use some of the tools I learned in my MBA class, like the **capital asset pricing model**, they did teach that back in the 80s by the way, so it's been around for a while. I think our cost of equity is around 10%"
>
> — CFO, Qorvo Inc. (Q4 2015)

> "And for those of us that took financial classes, undergrad or graduate, we all understand **CAPM** and WACC and so on and so forth, and we've done a good job, we believe, of managing our capital structure to minimize our cost of capital."
>
> — CFO, L Brands (Q4 2016)

> "This is our way of calculating the cost of equity. So this is very textbook like **CAPM** type of methodology. We start with the risk-free rate [...]. Then we look at the group's beta."
>
> — CFO, Talanx (Q4 2019)

> "If you don't push your ROE target up when interest rates go up, and that's for every company to decide, it depends if you believe in the **capital asset pricing model** or not. I won't take you back to school. [...] I think you should [expect a higher ROE]."
>
> — CFO, Intact Financial (Q3 2024)

> "So the investment is about $240 million. We expect to project finance the deal so we will be, I think, around $130 million equity investment. [...] I'm not going to get in to what our cost of capital is but most people can probably — I mean everybody knows **CAPM** and can kind of back into what it is."
>
> — CEO, Vistra Energy (Q1 2017)

> "Looking at WACC, we have both an equity component and a debt component. On the equity component, we calculate cost of equity using the **capital asset pricing model** using historical data and market risk premiums. Our current cost of equity is running around 9%."
>
> — CFO, Archer Daniels Midland (Q1 2012)

> "From a cost of capital perspective, we update our cost of capital estimates monthly. So yes, there are changes that go on a monthly basis looking at interest rates and looking at local country risk, as well as betas and all of the things that go into kind of the **CAPM** model."
>
> — CEO, American Tower Corp. (Q4 2013)

"We haven't historically disclosed our cost of equity, but it's actually fairly straightforward calculation from a **CAPM** model, so in terms of where we're coming out. But I think from that, you're going to get around a 10%-ish type range cost."

— CEO, E-Trade (Q1 2016)

"So the way I think about and we think about our cost of capital is, it's our long-term weighted average cost of capital. And with a balance sheet that's about a 75% equity, 25% debt spread. When you do a sort of a **capital asset pricing model** analysis of what is Camden's cost of capital, it's slightly higher than 6%."

— CEO, Camden (Q4 2019)

"Basically, we — across the portfolio, we apply hurdle rates based on market data and applying the usual **capital asset pricing model**, so we have specific targets."

— CFO, CLP Holdings (Q1 2022)

"We've also made progress on our weighted average cost of capital primarily through our refinancing, which resulted in about an 80-basis-point reduction in our weighted average cost of capital. In measuring that, we used kind of the classic **CAPM** type of calculation, which I'm sure we're all familiar with."

— CFO, Dynegy Inc. (Q1 2014)

"We estimate our cost of capital using a **CAPM** pricing model both on debt and equity, in a pretty traditional model. And we come up to about a 12% cost of capital."

— CEO, Eclipse Resources (Q1 2018)

"We've done analysis, and we've looked at a **CAPM** pricing model. We've looked at our WACC curve."

— Assistant Treasurer, Iron Mountain Inc. (Q2 2016)

"I know a year ago when I did the **capital asset pricing model** figure at our cost of equity it was about 13%."

— CEO, Pinnacle Entertainment (Q4 2007)

"[…] to determine the equity costs, there are no binding regulations or standards, but there is a number of different methodologies that can result in different results. Based on the so-called **capital asset pricing model**, we, at present, see our equity costs at group level at around 10% after tax."

— CFO, Deutsche Bank (AGM 2021)

"I mean we look at our cost of capital every quarter. We use **CAPM**. When we ran it at the balance sheet date it's about 6.2%"

— CFO, Great Portland (Q2 2019)

G Other Measures of Perceived Cost of Equity

This section outlines alternative measures of the perceived cost of equity. We use these measures to validate our main result that the perceived cost of capital rises with benchmarking. Specifically, we show that benchmarking-induced changes in a stock's CAPM $\hat{\beta}$ translate into higher perceived costs of equity reported by stock analysts and by regulators of public utilities and railroads.

G.1 Stock Analysts' Perceived Cost of Equity

We collect stock analysts' perceived cost of equity from three independent research providers: I/B/E/S, Morningstar, and Value Line. These firm sell their reports and advice to investors, creating an incentive to assign cost of equity that match investors' perceptions of a stock's risk. However, the providers use different methodologies to estimate the cost of equity which provides us with independent variation which we exploit to corroborate our main finding.

Morningstar Analysts' Cost of Equity We obtain Morningstar analysts' cost of equity directly from Morningstar Direct for the period from 2001 to 2018 for stocks in Morningstar's coverage universe listed on the NYSE, NASDAQ, and Amex. Morningstar's cost of equity consists of a common risk-free rate and a stock-specific risk premium, which reflects the stock's systematic risk as qualitatively assessed by an analyst. This approach means that cross-sectional variation in the cost of equity depends solely on Morningstar's perception of systematic risk. While Morningstar draws inspiration from the CAPM, it differs by using a qualitative, forward-looking assessment rather than simply applying the CAPM directly (for details see Morningstar, 2022, page 4f)).

Value Line Safety Rank We hand-collect and digitize Value Line Investment Survey reports for Small & Mid-Cap stocks from 1998 to 2006 to obtain Value Line's safety rank measure, using the last available rank in each calendar year. The safety rank, ranging from 1 (safest) to 5 (riskiest), reflects Value Line analysts' subjective assessment of a stock's price stability and the financial strength of the underlying firm. Jensen (2024) shows that the CAPM best describes the subjective risk assessment of Value Line (see also Brav et al., 2005).

In the main text, we use the safety rank as a proxy for the perceived cost of equity and follow Eskildsen et al. (2024) in converting the ordinal rank to a required return on equity by multiplying it by 1.5 p.p.. We show below that instead working directly with the original ordinal rank yields qualitatively similar results.

Table G11 reports the marginal effects from an ordered logit regression of the Value Line

Table G11: Change in Probability of Each Value Line Safety Rank in Response to Δ BMI = 10 p.p.

	Safe		Average	Risky	
Value Line Safety Rank	1	2	3	4	5
Δ BMI = 10 p.p.	-1.52***	-4.00***	-8.48***	11.77***	2.23***
	(0.35)	(1.10)	(1.71)	(2.37)	(0.68)
Baseline Probability	2.7%	8.0%	48.3%	37.0%	4.0%
Observations	2,524				
Brant-Test p-value	0.61				

Notes: This table reports marginal effects of an ordered logit regression of Value Line safety rank on changes in benchmarking intensity due to Russell index reconstitution. We restrict the sample to stocks within 400 ranks around the Russell index cutoffs. Standard errors in parentheses are clustered by year. * p<0.10, ** p<0.05, *** p<0.01.

safety rank on exogenous changes in benchmarking intensity due to Russell reconstitution. We restrict the sample to stocks within 400 ranks around the Russell index cutoffs. The coefficients indicate the change in the probability of each outcome category due to a 10 p.p. increase in BMI (from May to June) because of Russell index reconstitution. The results show that an exogenous increase in BMI lead to a significant increase in the Value Line safety rank. The probability that a stock's riskiness is classified as above average increases by more than 11 p.p. at benchmark inclusion. This suggests that Value Line's stock analysts perceive an increase in the required rate of return on equity when benchmarking intensity increases.

I/B/E/S Stock Analysts' Subjective Expected Returns I/B/E/S does not directly provide cost of equity estimates. However, we can infer stock analysts' perceived cost of equity from their subjective expected returns. To do this we use data on the consensus forecasts of stock analysts from I/B/E/S for the period from 2002 to 2018. We construct stock analysts' subjective expected returns from I/B/E/S as

$$\mathbb{E}_t^\star \left[R_{i,t+1}\right] = \frac{\mathbb{E}_t^\star \left[p_{i,t+1}\right] + \mathbb{E}_t^\star \left[d_{i,t+1}\right]}{p_{i,t}} \tag{16}$$

in which $\mathbb{E}_t^\star \left[p_{i,t+1}\right]$ and $\mathbb{E}_t^\star \left[d_{i,t+1}\right]$ are the median consensus one-year price target and dividend forecast over the next fiscal year, respectively, and $p_{i,t}$ is the stock's price at the day of the forecast from CRSP. The subjective expected returns constructed in Eq. (16) are based on analysts' forecasts of future stock prices and thus incorporate both perceived discount rates and perceived mispricing, that is, whether analysts think the stock is over- or undervalued (see Jensen, 2024).

G.2 Authorized Cost of Equity of Regulated Monopolies

This subsection provides further evidence that increased benchmarking affects the perceived cost of equity. Our analysis, which draws on Kontz (2025), uses comprehensive data on requested and authorized costs of capital for all major U.S. investor-owned utilities from 1998 to 2018, covering over three-quarters of U.S. consumers. Kontz (2025) analyzes how the growth of index investing impacts regulated monopolies' cost of equity and consumer energy prices. Specifically, we test whether regulators of monopolists perceive a higher cost of equity when a firms' CAPM $\hat{\beta}$ increases due to benchmarking.

Background Electricity and natural gas utilities operate as regulated monopolies, granted geographic exclusivity in exchange for rate oversight by government utility commissions. Because these utilities do not face market-based pricing, regulators use a cost-of-service approach: they evaluate the utility's costs and investments, assess their prudence, and apply a risk-adjusted return to determine the revenue requirement that sets customer rates. A central regulatory challenge is setting a fair return on equity (RoE). The legislative basis for this is U.S. Supreme Court (1944) in *Federal Power Commission v. Hope Natural Gas Co.* which ruled that a regulated monopoly's "[...] return to the equity owner should be commensurate with returns on investments in other enterprises having corresponding risks." Today, state and federal regulators usually implement the CAPM or a version of the DCF model to estimate the cost of equity.

Public Utilities We study utility rate cases from 1998 to 2018, covering all major investor-owned electricity and natural gas utilities in the U.S., which collectively serve over three-quarters of U.S. consumers. We collect data on requested and authorized costs of equity (CoE) from Regulatory Research Associates. We test whether benchmarking affects the authorized CoE using IV specifications of the following form:

$$\text{Authorized CoE}_{i,t} - r_t^f = \alpha_i + \lambda \, \widehat{\text{CAPM } \beta}_{i,t} + \varphi \left(\text{DCF Implied CoE } - r^f \right) + \varepsilon_{i,t}, \tag{17}$$

in which $\widehat{\text{CAPM } \beta}_{i,t}$ is instrumented with the firm's benchmarking intensity, and the DCF-implied CoE term enters as a control. We include utility-by-state fixed effects, α_i, which absorb time-invariant unobserved heterogeneity across utility-state pairs. Identification of λ and φ thus relies on within-utility time-series variation.

Table G12 reports coefficient estimates of Eq. (17). Columns 1 and 2 show that a higher benchmarking intensity results in a higher authorized cost of equity. A 10 p.p. higher benchmarking intensity translates into a 70 bps higher authorized cost of equity. Columns 3 and 4 translate

Table G12: Regulated Monopolies' Cost of Equity and Benchmarking Intensity

	Dependent Variable: Authorized Cost of Equity $- r^f$							
	Public Utilities				Railroads			
	RF		IV		RF		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
BMI (in %)	0.069***	0.071***			0.481***	0.454***		
	(0.011)	(0.011)			(0.099)	(0.088)		
$\widehat{\text{CAPM } \beta^E}$			6.064***	6.189***			6.462***	6.481***
			(1.386)	(1.375)			(0.331)	(0.361)
DCF implied Cost of Equity $- r^f$		0.281***		0.170**		0.654**		-0.035
		(0.051)		(0.083)		(0.327)		(0.072)
Fixed Effects								
Utility × State FE	✓	✓	✓	✓				
Adj. R^2	0.26	0.43			0.48	0.60		
FS F-stat.			42.5	45.4			23.1	25.4
Observations	1,052	1,052	1,052	1,052	21	21	21	21

Notes: This table reports coefficient estimates of the form: Authorized $\text{CoE}_{i,t} - r_t^f = \alpha_i + \lambda \widehat{\text{CAPM } \beta}_{i,t} + \varphi \left(\text{DCF Implied CoE } - r^f\right) + \varepsilon_{i,t}$ for rate regulated public utilities' and railroads' authorized cost of equity. Data for authorized cost of equity for public utilities and railroads are from Regulatory Research Associates and from the Surface Transportation Board, respectively. CAPM $\hat{\beta}$ estimated from weekly returns data as usual in regulatory proceedings. Standard errors in parentheses clustered at utility-level for public utilities and Newey-West with 5 lags for railroads. * p<0.10, ** p<0.05, *** p<0.01

the reduced form coefficient into the perceived CAPM-implied equity risk premium by instrumenting $\hat{\beta}$ with BMI. The IV estimates imply a risk premium of around 6.1%. Our estimate is close to the historical equity risk premium observed in the U.S. which is often used in regulatory proceedings.

Regulators often require analysts to estimate a public utility's cost of equity using a discounted cash flow (DCF) method in addition to the CAPM. The even-numbered columns in Table G12 show that our results remain robust when accounting for the DCF-implied risk premium. While the DCF risk premium explains a large share of the variation in authorized risk premia, its inclusion has only a negligible effect on the BMI coefficient. This suggests that omitted variable bias is unlikely to be a concern (Oster, 2017).

Appendix Table G13 shows that benchmarking intensity does not correlate with the authorized cost of debt of public utilities. In contrast, controls for aggregate credit market conditions, such as the BBB option-adjusted spread, exhibit a highly significant correlation with both requested and authorized cost of debt. This provides confidence that BMI serves as a valid instrument for the cost of equity by influencing CAPM $\hat{\beta}$s while not affecting the cost of capital through other channels.

Table G13: Effect of Benchmarking on Utilities Requested and Authorized Return on Debt

	(1)	(2)	(3)	(4)	(5)	(6)
	Requested			Authorized		
	Return on Debt $-$ R^f			Return on Debt $-$ R^f		
Benchmarking Intensity (in %)	0.016	0.008	0.009	-0.002	-0.013	-0.020
	(0.012)	(0.011)	(0.011)	(0.014)	(0.013)	(0.015)
BBB Option-Adjusted spread		0.264***	0.259***		0.310***	0.296***
		(0.048)	(0.049)		(0.053)	(0.057)
Requested E/(D+E)		0.075***	0.071***			
		(0.016)	(0.018)			
Authorized E/(D+E)					0.072***	0.066***
					(0.018)	(0.016)
Constant	2.218***	-1.979*		3.502***	-0.487	
	(0.267)	(0.753)		(0.295)	(0.878)	
Utility-by-State Fixed Effect			✓			✓
Adj. R^2	0.00	0.14	0.38	0.00	0.13	0.39
Observations	1,381	1,381	1,347	1,022	1,022	987

Notes: This table shows coefficient estimates for Return on $\text{Debt}_{i,t} = \alpha_i + \text{BMI}_{i,t} + \xi X'_{i,t} + \nu_{i,t}$ Risk-free rate (R^f_t) is the nominal yield on 10-year Treasurys. Standard errors clustered at utility and year-quarter in parenthesis. * p<0.10, ** p<0.05, *** p<0.01.

Railroads We use data on the cost of equity for regulated railroads from the Surface Transportation Board (STB). The STB sets an industry-wide annual cost of equity, rather than firm-specific rates. We thus only have a limited number of yearly observations. However, the STB data offers a granular view of the regulatory rate-setting process: the STB reports the risk-free rate, CAPM $\hat{\beta}$, and equity risk premium used to determine the industry-wide cost of equity. Importantly, the STB's equity risk premium enables us to assess the accuracy of our IV-implied estimates. We combine the STB data with the average BMI of publicly traded railroad companies.

Columns (5) to (8) of Table G12 report results for the railroad industry. Benchmarking intensity strongly predicts the authorized cost of equity, even after controlling for the DCF-implied cost. The IV estimates imply a perceived CAPM equity risk premium of 6.4% annually—statistically indistinguishable from the average 6.85% applied by the STB over the sample period.

H Additional Tests and Instrument Validity

H.1 Changes in BMI and Measures of Risk Exposure

Changes in BMI that correlate with changes in exposure to aggregate or idiosyncratic risk pose a threat to our identification strategy. Industry's exposure to aggregate risk (Karolyi, 1992) and firm fundamentals (Gomes et al., 2003) should in theory determine firm-level exposure to aggregate risk. We thus test whether the aggregate risk exposure of treated firms changes by estimating whether the CAPM $\hat{\beta}$ of comparable peer firms changes when a firm's BMI changes. We also test whether measures of idiosyncratic firm-level risk exposure change with BMI. However, we find no evidence that changes in BMI correlate with changes in risk exposure.

Data We collect peer group information from data on executive compensation by Institutional Shareholder Services (ISS). We use measures of firm-level intangible capital from Peters & Taylor (2017). We use firm-level risk measures from Hassan et al. (2019) and data on financial frictions from Hoberg & Maksimovic (2015) and Linn & Weagley (2024).

Table H14: Placebo test using the CAPM $\hat{\beta}$ of peer firms

	(1) Firm's Δ CAPM β^E	(2) Firm's Δ CAPM β^E	(3) All Peers' Δ CAPM β^E	(4) Peer n=1 Δ CAPM β^E	(5) Peer n=2 Δ CAPM β^E	(6) Peer n=3 Δ CAPM β^E
Δ CAPM β^{Peer}	0.218***					
	(0.027)					
Δ BMI		0.511**	0.042	0.064	0.085	-0.023
		(0.176)	(0.053)	(0.092)	(0.081)	(0.083)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.20	0.30	0.13	0.12	0.12	0.14
Observations	16,021	5,302	16,416	5,641	5,514	5,261

Notes: This table reports coefficient estimates for a placebo test using N=3 firm peers' change in CAPM $\hat{\beta}$ and assigns them the Δ BMI of the firm: $\Delta\text{CAPM } \beta^{\text{Peer}}_{j,t} = \alpha_t + \Delta\text{BMI}^{\text{Firm}}_{i,t} + \log(\text{Market Cap.}) + \varepsilon_{j,i,t}$ for firm i and peer j in year t. Standard errors in parentheses clustered at the firm- and year-level in column (1) and (2) and clustered at firm-, peer-, and year-level in other columns. Estimated within 300 ranks around the Russell 1000/2000 cutoff. * p<0.10, ** p<0.05, *** p<0.01

Changes in CAPM $\hat{\beta}$s of Peer Firms We collect information about a firm's peer group from ISS. For each firm, we randomly select three peer firms and test whether the firm's change in BMI correlates with changes in the CAPM $\hat{\beta}$ of peers. To avoid confounding our estimates, we exclude peers that also experience a change in BMI. Appendix Table H14 reports the peer-based test. Column (1) shows that regressing changes in a firm's CAPM $\hat{\beta}$ on changes in its peers'

$\hat{\beta}$s yields a significant positive coefficient, consistent with common exposure to aggregate risk.[10] Column (2) shows that changes in a firm's BMI change the firm's $\hat{\beta}$. By contrast, changes in a firm's BMI do not correlate with changes in its peers' $\hat{\beta}$s, with insignificant coefficients near zero. This pattern suggests that benchmarking changes a firm's CAPM $\hat{\beta}$ rather than reflecting changes in aggregate risk exposure.

Table H15: Changes in CAPM $\hat{\beta}$ and firm-level risk measures of Hassan et al. (2019)

(in σ units)	(1) Δ Risk	(2) Δ P. Risk	(3) Δ P. Risk Economic	(4) Δ P. Risk Security	(5) Δ P. Risk Technology	(6) Δ P. Risk Trade
Δ BMI	-0.020	-0.006	-0.003	0.003	-0.009	-0.004
	(0.014)	(0.014)	(0.012)	(0.011)	(0.011)	(0.016)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.03	0.01	0.02	0.01	0.01	0.02
Observations	9,368	9,264	9,190	9,138	9,093	9,093

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Firm-level Risk}_{i,t} = \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$. Changes in firm-level risk (Hassan et al., 2019) calculated between 1st and 4th quarter of the year. Coefficients are standardized to unit variances. Changes in firm-level risk measures are trimmed at the 1% and 99% level. Estimated within 300 ranks around the Russell 1000/2000 cutoff. Standard error in parentheses are clustered at the firm- and year-level. * p<0.10, ** p<0.05, *** p<0.01

Firm-level Risk Measures We analyze six firm-level risk measures derived from earnings calls by Hassan et al. (2019): the overall risk exposure of firms, exposure to overall political risk, and exposure to political risk stemming from economic, security, technological, or trade policy. Appendix Table H15 reports estimates of OLS regressions of changes in firm-level risk measures on changes in BMI. Changes in BMI do not correlate with changes in firm-level risk measures. The estimated coefficients across all risk measures are close to zero and statistically insignificant.

H.2 Changes in BMI and Measures of Financial Constraints

Changes in BMI could correlate with changes in financial constraints, potentially violating the exclusion restriction of our IV strategy. We test this by examining the correlation between changes in a firm's BMI and measures of financial constraints and CDS spreads. If changes in BMI correlated with changes in financing costs due to factors other than CAPM $\hat{\beta}$, the exclusion restriction would be violated. However, we find no evidence of such correlations.

Text-based Measures of Financial Constraints We collect text-based measures of financial constraints from Hoberg & Maksimovic (2015) and Linn & Weagley (2024). These measures capture the extent to which firms face financial constraints and are likely to constrain investment

[10] Levi & Welch (2017) similarly show that peer firms' CAPM $\hat{\beta}$s strongly predict own-firm $\hat{\beta}$s.

Table H16: Changes in measures of text-based financial frictions (Hoberg & Maksimovic, 2015)

(in σ units)	(1) Δ Inv. Delay	(2) Δ Inv. Delay & Equity Issue	(3) Δ Inv. Delay & Debt Issue	(4) Δ Inv. Delay & Private Issue	(5) Δ Inv. Delay & Equity (LW, '24)	(6) Δ Inv. Delay & Debt (LW, '24)
Δ BMI	-0.011	0.000	-0.017	-0.011	-0.010	-0.018
	(0.008)	(0.010)	(0.019)	(0.013)	(0.011)	(0.013)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.01	0.01	0.02	0.00	0.01	0.01
Observations	5,688	5,686	5,669	5,713	7,674	7,624

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Measure of Financial Constraint}_{i,t} = \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$. Changes in text-based financial constraint measures from Hoberg & Maksimovic (2015) and Linn & Weagley (2024). Coefficients are standardized to unit variances. Changes in financial constraints measures are trimmed at the 1% and 99% level. Standard error in parentheses are clustered at the firm- and year-level. Estimated within 300 ranks around the Russell 1000/2000 cutoff. * p<0.10, ** p<0.05, *** p<0.01.

based on the text of their annual reports. Appendix Table H16 reports estimates of OLS regressions of changes in the firm's financial constraints on changes in the BMI of a firm. The estimated coefficients of BMI are close to zero and not statistically significant across all measures. Importantly, Column (1) of Appendix Table H16 shows that changes in BMI do not correlate with firm statements about plans to delay investments.

Table H17: Changes in CDS Spreads and CAPM $\hat{\beta}$s of CDS Spreads

	(1) Δ CDS Spread	(2) Δ CDS Spread	(3) Δ CDS Beta	(4) Δ CDS Beta
Δ BMI	-0.054	-0.063	-0.020	-0.031
	(0.056)	(0.049)	(0.040)	(0.032)
Momentum (Cum. Ret.)		-0.188***		-0.090
		(0.043)		(0.059)
Log(Market Cap.)	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Adj. R^2	0.14	0.16	0.00	0.01
Observations	781	781	614	614

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{CDS Spreads}_{i,t} = \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$. Coefficients are standardized to unit variances. CDS spreads for senior unsecured debt with tenor of 5 year and doc clause XR14 (no restructuring). CDS CAPM $\hat{\beta}$s are calculated on daily data from 2010 to 2019 using the weighted least squares estimator of Welch (2022) with exponentially decay of 3 months half life. Changes in CDS spreads and CDS CAPM $\hat{\beta}$s are winsorized at the 1% and 99% level. Standard error in parentheses are clustered at the firm- and year-level. Estimated within 300 ranks around the Russell 1000/2000 cutoff. * p<0.10, ** p<0.05, *** p<0.01.

Changes in CDS Spreads and CDS CAPM $\hat{\beta}$ We collect CDS spreads for senior unsecured debt with tenor of 5 year from 2010[11] to 2019. We calculate CDS CAPM $\hat{\beta}$s on daily data using the estimator of Welch (2022). We calculate changes in a firm's CDS spreads and firm's CAPM $\hat{\beta}$ of CDS spreads as the difference between the average of daily observations in the first and last

[11]We focus on the period after ISDA's "Big Bang" reforms of April 2009 to maintain a consistent sample.

quarter of a year. Appendix Table H17 reports estimates of OLS regressions of changes in CDS spreads and changes in the CDS CAPM $\hat{\beta}$ on changes in the BMI of a firm. We find no evidence that changes in the BMI predict changes in firm CDS spreads or CDS CAPM $\hat{\beta}$s. The estimated coefficients on BMI are insignificant and close to zero.

Table H18: Changes in measures of corporate governance

(in σ units)	(1) Δ SP G-Score	(2) Δ Sus. G-Score	(3) Δ Ref G-Score	(4) Δ SP ESG	(5) Δ Sus. ESG	(6) Δ Ref. ESG
Δ BMI	-0.066	0.004	0.006	-0.033	0.005	-0.016
	(0.072)	(0.023)	(0.020)	(0.059)	(0.019)	(0.018)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.03	0.09	0.03	0.04	0.17	0.05
Observations	419	2,370	4,123	417	2,449	4,137

Notes: This table reports coefficients estimates for regression specifications of the form: $\Delta\text{Governance Score}_{i,t} = \alpha_t + \gamma\,\Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$. Governance and ESG scores of Standard & Poor, Sustainalytics, and Refinitiv. Coefficients are standardized to unit variances. Changes in G-Scores and ESG Scores are trimmed at the 1% and 99% level. Standard error in parentheses are clustered at the firm- and year-level. Estimated within 300 ranks around the Russell 1000/2000 cutoff. * p<0.10, ** p<0.05, *** p<0.01.

H.3 Changes in BMI and Measures of Corporate Governance

An increase in BMI and associated institutional ownership could impact investment through improved corporate governance (Appel et al., 2016, Aghion et al., 2013). However, increased passive ownership may also decrease monitoring incentives, as in the model of Bebchuk & Hirst (2019). We test whether measures of governance change with changes in BMI but find no evidence of such an effect.

We use governance and ESG scores from S&P, Sustainalytics, and Refinitiv and test whether changes in BMI correlate with changes in those scores. Appendix Table H18 reports estimates of OLS regressions of changes in governance and ESG scores on changes in the BMI of a firm. The estimated coefficients are close to zero and are not statistically significant. Our findings are consistent with Kacperczyk et al. (2021), who also find no evidence of changes in governance at benchmark inclusion.

Activist Investor Campaigns An alternative channel through which benchmarking could affect investment is by altering the incentives for activist investor campaigns. A BMI could increase the likelihood of an activist campaign through two mechanisms. First, higher BMI increases a stock's liquidity, which reduces the cost for an activist to accumulate a large stake. Second, a larger passive investor base, which correlates with BMI, may increase an activist's probability of success in a proxy contest (Appel et al., 2019, Kedia et al., 2021). However, countervailing

economic forces also exist. An activist targeting a high-BMI firm must bear the higher factor risk associated with the increases in stock's CAPM $\hat{\beta}$. This factor risk increases the cost and uncertainty of an activist campaign.

Table H19: Marginal Effects of BMI on the Probability of an Activist Investor Campaign

Outcome y	ΔPr(y) when ΔBMI=10 p.p.	SE	p-value	No. Obs.	Mean y	# y=1
Activist Campaign (12 mos.)	-0.0017	0.0007	0.02	18,059	0.0156	282
Activist Campaign (24 mos.)	-0.0000	0.0008	0.95	18,059	0.0246	444
Activist Campaign (36 mos.)	-0.0004	0.0015	0.76	18,059	0.0328	592
Activist Campaign (48 mos.)	0.0005	0.0018	0.76	18,728	0.0408	765
Activist Campaign (60 mos.)	0.0003	0.0020	0.86	18,728	0.0479	898

To test this channel, we use data on activist campaigns from FactSet's SharkWatch, restricting the sample to campaigns with the objective to "Maximize Shareholder Value." We estimate the following logit model:

$$\Pr\left[\text{Activist Campaign}_{i,t+h} = 1\right] = \Lambda\left(\alpha_t + \gamma\,\Delta\text{BMI}_{i,t} + X'_{i,t}\xi + \nu_{i,t+h}\right)$$

in which $\Lambda(\cdot)$ is the logistic function. The dependent variable is an indicator for an activist campaign initiated against firm i between month t and $t+h$. We test horizons $h \in \{12, 24, 36, 48, 60\}$ following the Russell reconstitution. The vector $X'_{i,t}$ contains the log market capitalization and banding controls.

Table H19 reports marginal results for a 10 p.p. increase in BMI. We find no evidence that an increase in BMI leads to a higher probability of an activist campaign over a one- to five-year horizon. The coefficient on ΔBMI is statistically indistinguishable from zero for all horizons of 24 months or longer. For the 12-month horizon, the coefficient is negative, pointing to a decrease in the probability of an activist campaign. These findings do not support the hypothesis that activism is a channel through which benchmarking affects firm investment.

I Semi-Elasticity of Investment by Sales in Compustat

Our analysis is similar to Zwick & Mahon (2017), who study the bonus depreciation stimulus policies from 2001 to 2004 and from 2008 to 2010. In both episodes, firms were allowed to use an accelerated schedule to deduct the cost of investment purchases from taxable income, thereby affecting the relative price of capital q_t. We depart by focusing on the cost of equity part of the user cost of capital, $C_t = q_t \left[(1-\mu_t)(1-\tau_t)r_t^d + \mu_t(r_t^f + \beta_{i,t}\lambda) + \delta\right]$, which benchmarking affects by changing CAPM $\hat{\beta}$s.

We estimate the investment semi-elasticity with respect to the user cost of capital (ϵ^C) for each of 10 sales deciles. Figure I15 plots our OLS estimates of the semi-elasticities by sales decile. The figure shows that smaller firms, as measured by sales, have more negative investment semi-elasticities than larger firms. The semi-elasticity estimates for the smallest sales decile is -9, while that for the largest sales decile is around -1. The difference between the smallest and largest sales decile is statistically significant at the 1% level. The estimates are robust to adding controls for cash flow, operating leverage (SGA/sales), and log MRPK (blue circles) and further adding industry by time fixed effects (green diamonds). The results are consistent with prior research showing that smaller firms' investment is more sensitive to changes in the user cost of capital than larger firms' (e.g., Zwick & Mahon, 2017).

To estimate the investment semi-elasticity with respect to the user cost of capital (ϵ^C) for each of 10 sales deciles, we use Compustat data from 1989 to 2019 in a three-step procedure.

1. We first estimate the elasticity of investment with respect to CAPM $\hat{\beta}$ (ε_j^β) by regressing the change in the log investment-to-capital ratio on log CAPM beta, including firm and year fixed effects and controls ($X_{i,j,t}$):

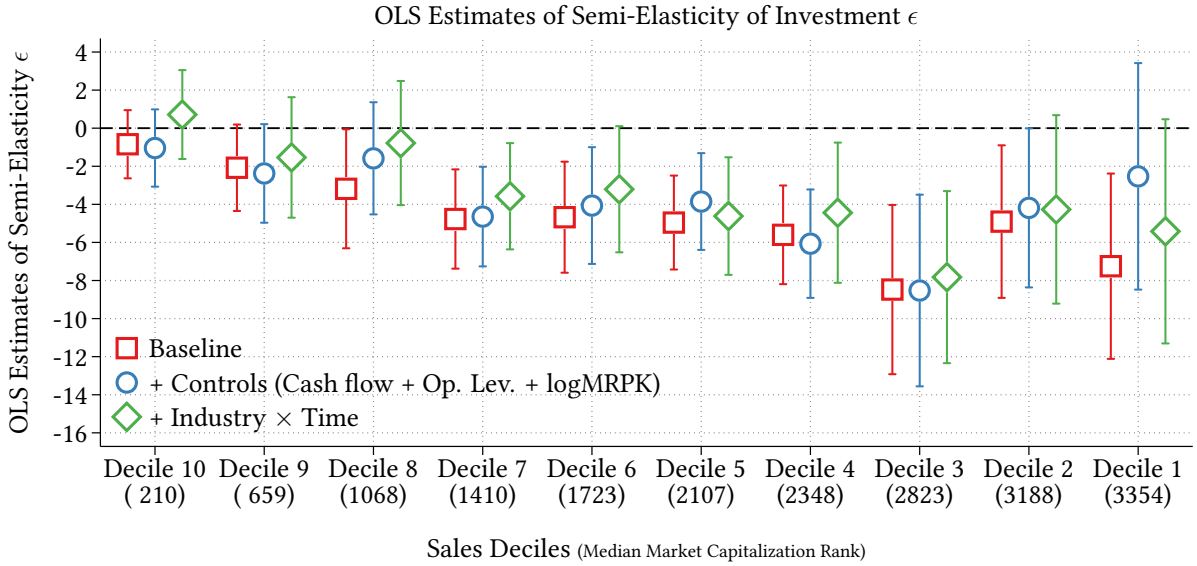
$$\Delta \log(I/K)_{i,j,t} = \alpha_i + \alpha_t + \varepsilon_j^\beta \log(\hat{\beta}_{i,j,t}) + X'_{i,j,t}\zeta + \varepsilon_{i,j,t} \tag{18}$$

2. We then convert the estimated elasticity (ε_j^β) to the target semi-elasticity (ϵ_j^C) using:

$$\epsilon_j^C = \frac{\partial \log(I/K)}{\partial C_j} = \underbrace{\frac{\partial \log(I/K)}{\partial \log \beta_j}}_{\text{Step 1: } \widehat{\varepsilon}_j^\beta} \times \underbrace{\frac{1}{\bar{\beta}_j}}_{\text{Elasticity to Semi-Elasticity}} \times \underbrace{\left(\frac{\partial C_j}{\partial \beta_j}\right)^{-1}}_{\text{Rescaling to User Cost}}$$

3. The final term, the sensitivity of the user cost to β, is $(\partial C/\partial\beta)^{-1} = (q\mu\lambda)^{-1}$. We use the average equity share for each decile (μ_j) and assume an equity risk premium (λ) of 6% and

Figure I15: Semi-Elasticity of Investment by Sales Deciles



Notes: The figure plots estimates of the semi-elasticity of investment with respect to the user cost of capital, ϵ^C, by sales decile. Numbers in parentheses on the x-axis report the median May market-capitalization rank for each sales decile. Red squares show the baseline specification with firm and year fixed effects. Blue circles add controls for cash flow, operating leverage (SGA/sales), and log MRPK. Green diamonds further add industry by time fixed effects. 95% confidence intervals based on standard errors double-clustered by firm and year.

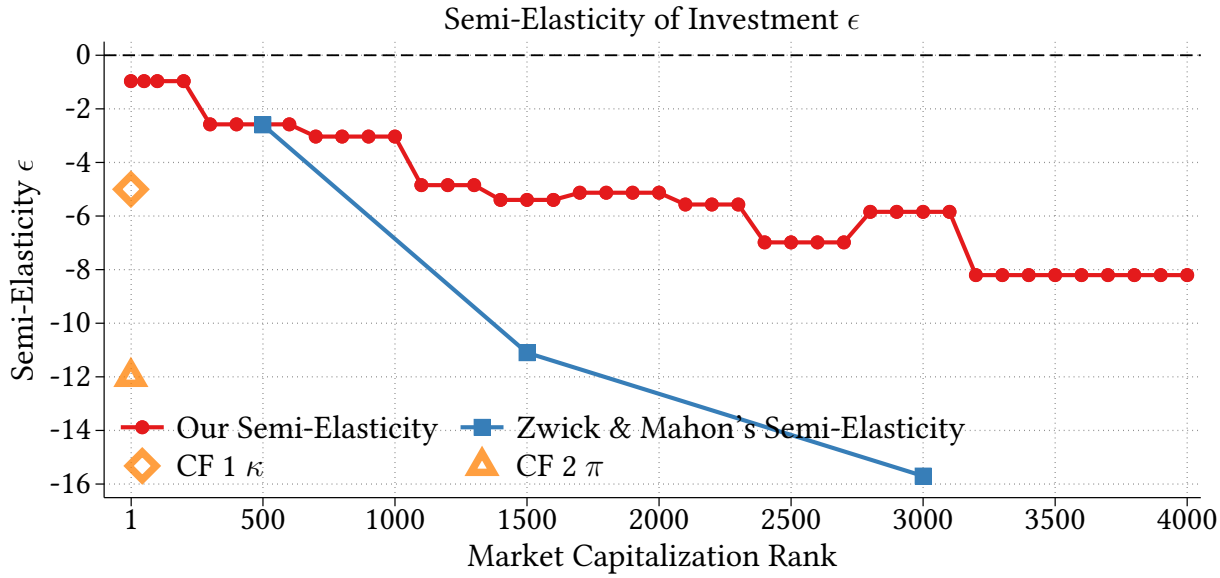
a relative price of capital (q) of 1.

Attenuation Bias in OLS Estimates Our estimates of the investment semi-elasticity with respect to the user cost of capital (ϵ^C) are likely biased towards zero for several reasons. First, our CAPM $\hat{\beta}$ are subject to classical measurement error, which attenuates the coefficient toward zero. Second, investment-opportunity shocks might confound our estimates. For instance, a positive productivity shock could raise both investment and $\hat{\beta}$ (via operating leverage), masking the true negative relationship. Third, an expectations channel may violate our identification strategy: if managers, similar to analysts raise growth expectations (Jylha & Ungeheuer, 2021), when β increases, investment may increase for reasons not running through the user cost, again biasing our coefficient toward zero.

Recognizing these biases, we compare our OLS estimates to the well-identified estimates from Zwick & Mahon (2017). Their tax-based identification is less susceptible to the confounders we face, and in a standard investment model, the elasticity with respect to the user cost is proportional to the elasticity with respect to the net-of-tax rate (Koby & Wolf, 2020).

Figure I16 plots our scaled estimates and estimates from Zwick & Mahon (2017) by market cap-

Figure I16: Semi-Elasticity of Investment by Market Capitalization Rank



Notes: The figure plots estimates of the semi-elasticity of investment with respect to the user cost of capital, ϵ^C, by market-capitalization rank. We map our sales-decile estimates to ranks using the median market-cap rank within each decile. For comparison, we overlay estimates from Zwick & Mahon (2017), positioned at the median rank of their small- and large-firm groups. Orange markers (CF1 κ and CF2 π) show the semi-elasticities that the largest 50 firms would need to offset the observed decline in aggregate investment implied by the cross-sectional rise in CAPM $\widehat{\beta}$; the two markers use alternative weighting schemes.

italization rank, confirming our main result: smaller firms have more negative investment semi-elasticity than larger firms. Zwick & Mahon (2017) find an average investment semi-elasticity with respect to the user cost of approximately -11, which is twice the size of our unadjusted estimates.[12]

[12]The estimates from Zwick & Mahon (2017) for the Compustat sample are taken from their Appendix Table B.2. The estimates need to be rescaled by the investment tax rate to represent the semi-elasticity with respect to the user cost of capital, as described on their page 230 (see also Koby & Wolf, 2020, p.21f).

J Aggregation using a Heterogeneous Firm General Equilibrium Model

Set-up Our discrete-time model builds closely on Khan & Thomas (2008) and Winberry (2021).

Firms There is a fixed mass of firms $j \in [0,1]$ producing output $y_{j,t}$ with technology

$$y_{j,t} = z_t \varepsilon_{j,t} k_{j,t}^{\theta} l_{j,t}^{\nu},$$

in which z_t is an aggregate productivity shock, $\varepsilon_{j,t}$ is an idiosyncratic productivity shock, $k_{i,t}$ is the capital stock, and $l_{j,t}$ is labor input. The parameters θ and ν are the output elasticities of capital and labor, respectively. Throughout, we assume that $\theta + \nu < 1$.

The aggregate shock process is common across all firms while the idiosyncratic shock $\varepsilon_{j,t}$ is independent across firms. The two shocks follow AR(1) processes in logs:

$$\begin{aligned} \log z_{t+1} &= \rho_z \log z_t + \sigma_z \epsilon_{z,t+1}, \quad \epsilon_{z,t+1} \sim N(0,1), \\ \log \varepsilon_{j,t+1} &= \rho_\varepsilon \log \varepsilon_{j,t} + \sigma_\varepsilon \epsilon_{\varepsilon,j,t+1}, \quad \epsilon_{\varepsilon,j,t+1} \sim N(0,1). \end{aligned}$$

Each period, a firm j observes these two shocks, uses its pre-existing capital, hires labor from a competitive labor market at wage w, and produces output. After production, the firm decides how much it should investment for the next period. Capital follows the usual law of motion $k_{j,t+1} = (1-\delta)k_{jt} + i_{jt}$, in which δ is the depreciation rate and i_{jt} is investment.

Firms face two types of capital adjustment costs. First, there is a fixed investment cost, $\xi_{j,t} \overset{iid}{\sim}$ uniform$[0, \bar{\xi}]$, paid in labor units if the firm invests a positive amount. Second, there is a standard convex investment adjustment cost. The adjustment cost function takes the form

$$\phi(k_{j,t}, i_{j,t}, \xi_{j,t}; \kappa) = \frac{\kappa}{2} \left(\frac{i_{jt}}{k_{jt}} \right)^2 k_{jt} + \xi_{jt} w(\boldsymbol{s}) \mathbb{1}\{i_{jt} > 0\}.$$

Households There is a representative household with GHH preferences over consumption C_t and labor supply L_t, which maximizes their expected discounted lifetime utility

$$\begin{aligned} &\max_{C,N} E_0 \sum_{t=0}^{\infty} \rho^t \log \left(C_t - \chi \frac{L_t^{1+\eta}}{1+\eta} \right) \\ &\text{subject to} \quad C_t = w_t L_t + \Pi_t \end{aligned}$$

in which ρ is the discount factor, χ is the weight on disutility from labor, and η is the inverse of the Frisch elasticity of labor supply. We normalize the total time endowment to 1, so that $L_t \in [0, 1]$. Markets are complete, the household owns all firms and rents labor to them in a competitive labor market at wage w_t.

Recursive firm problem The firms' individual state variables are its current draw of the idiosyncratic productivity shock, ε, its capital stock, k, and its investment fixed cost shock, ξ. The aggregate state is $\boldsymbol{s} = (z, \mu)$ where z is the aggregate productivity shock and μ is the distribution of firms over their individual states (ε, k, ξ). Note that that the firm's choice of labor is a purely static problem and we can thus separate it out to define the revenue function as

$$\pi(\varepsilon, k, \xi; \boldsymbol{s}) = \max_n \left\{ z\varepsilon k^\theta l^\nu - w(\boldsymbol{s}) l \right\}$$

The firm's value function takes the following form

$$v(\varepsilon, k, \xi; \boldsymbol{s}) = \pi(\varepsilon, k, \xi; \boldsymbol{s}) + \max_i \left\{ -q(\boldsymbol{s}) i - \phi(k, i, \xi; \kappa) + \mathbb{E}\left[\Lambda(z'; \boldsymbol{s})\, v(\varepsilon', k', \xi'; \boldsymbol{s}') | \varepsilon, k, \xi, \boldsymbol{s} \right] \right\}$$

where $\Lambda(z'; \boldsymbol{s})$ is the stochastic discount factor and q is the adjusted relative price of investment. Markets are complete with respect to aggregate risk, so the stochastic discount factor used by firms is equal to the household's intertemporal marginal rate of substitution state by state.

Equilibrium Definition and Solution Method The aggregate state vector is $\boldsymbol{s} = (z, \mu)$, where z is aggregate productivity and μ is the distribution of firms' individual state vector (ε, k, ξ). The formal equilibrium definition and solution concept is given in Winberry (2021).

J.1 Calibration

We parameterize the model in two steps. First, we calibrate a subset of parameters to match standard steady-state macroeconomic targets. Second, conditional on these values, we calibrate the remaining parameters to match data moments. One model period equals one quarter.

Fixed Parameters Panel B of Table J20 summarizes our fixed parameters. We mostly follow Winberry (2021) and set the household discount factor to $\rho = 0.99$, the inverse of the Frisch elasticity to $\eta = 0.50$, the labor share to $\theta = 0.64$, the capital share to $\nu = 0.21$, and the depreciation rate to $\delta = 0.03$. We set the aggregate TFP process to $\rho_z = 0.97$ and $\sigma_z = 0.008$.

Calibrated Investment Parameters We calibrate the parameters governing the firm heterogeneity in panel B of Table J20 such that the model matches key micro investment moments in

Table J20: Micro investment moments and calibration

Panel A: Target moments

Moment	Data	Model
Investment moments		
Average investment rate (%)	14.9	13.0
Standard dev. of investment rates (%)	15.6	13.9
Spike rate (%)	20.7	22.0
Perceive cost of equity moments		
Share of employment by top firms w/ lower $\hat{\beta}$ (in %)	33.5	32.6
Weighted-average increase in cost of capital (in %)	1.38	1.39

Panel B: Fixed & calibrated parameters

Block	Parameter	Value
Household (fixed)		
Discount rate	ρ	0.99
Inverse Frisch elasticity	η	0.50
Firm (fixed)		
Labor share	ν	0.64
Capital share	θ	0.21
Capital depreciation	δ	0.03
Aggregate TFP AR(1)	ρ_z	0.97
	σ_z	0.008
Firm heterogeneity (calibrated)		
Upper bound on fixed cost	$\bar{\xi}$	0.55
Convex adjustment cost	κ	3.10
Idiosyncratic productivity AR(1)	ρ_ε	0.90
	σ_ε	0.05
Perc. cost of equity heterogeneity (calibrated)		
Shape of $\hat{\beta}$	α	2.5
Upper bound of $\hat{\beta}$	$\bar{\beta}$	0.20
Lower bound of $\hat{\beta}$	$\underline{\beta}$	-0.058
Benchmarking shock AR(1)	ρ_λ	0.95

Notes: This table summarizes our calibration strategy. Panel A reports micro investment moments from the data and the model. Panel B reports fixed and calibrated parameters. Micro investment moments from our annual firm-level Compustat panel, 1998–2018. Statistics drawn from distribution of investment rates pooled over firms and time. Spike rate is fraction of observations with investment rate greater than 20 percent.

our Compustat panel shown in panel A of Table J20. We exogenously fix the persistence of the idiosyncratic productivity process to $\rho_\varepsilon = 0.90$.

Mapping our Empirical Findings to the Model We introduce shocks to firms' discount rates that depend on firm size to capture the empirical patterns we document. Rather than embedding a full asset pricing structure with subjective expectations, we model these shocks as wedges in the relative price of capital.[13] This preserves tractability while capturing the central mechanism in the data. Moreover, this is consistent with our empirical evidence that the change in $\hat{\beta}$ originates in the financial market and changes the *perceived* riskiness of equity while the fundamental risk-exposure is still determined by the firm's cash flows produced by $y_{i,t}$.

We model changes in CAPM $\hat{\beta}$ due to benchmarking as size-dependent, exogenous shocks to

[13]The relevant semi-elasticities between r and q are intimately linked as Proposition 4 in Koby & Wolf (2020) show.

the relative price of investment:

$$\widehat{q}(\boldsymbol{s}) = 1 + \lambda \times \left[\beta_0 + \beta_1 \exp\left(-\alpha\, n(\varepsilon, k; \boldsymbol{s})\right)\right]$$

in which β_0, β_1, and α are constants which we calibrate to match the empirical distribution of changes in firms' CAPM $\hat{\beta}$s. Specifically, $\beta_0 = \underline{\beta} - \frac{\bar{\beta}}{e^a - 1}$ and $\beta_1 = \bar{\beta}\frac{e^a}{e^a - 1}$ where $\underline{\beta}$ is the average decrease in CAPM $\hat{\beta}$ experienced by the largest firms, $\bar{\beta}$ is average increase in CAPM $\hat{\beta}$ experienced by smaller firms, and α governs how quickly the change in CAPM $\hat{\beta}$ decreases with firm size. That is, the minority of large firms in the model experience a decrease in their discount rate, while the majority of smaller firms experience an increase. We assume that the benchmarking shock λ follows an AR(1) process

$$\lambda_{t+1} = \rho_\lambda \lambda_t + \epsilon_{\lambda,t+1}, \quad \epsilon_{\lambda,t+1} \sim N(0, 1).$$

We set the quarterly autocorrelation of $\rho_\lambda = 0.95$ to match the persistent of CAPM $\hat{\beta}$s.

We calibrate α, $\bar{\beta}$, and $\underline{\beta}$ to match two key empirical moments and the distribution of changes in CAPM $\hat{\beta}$s documented in Section 3. Specifically, we target the average share of employment held by firms with a decrease in their CAPM $\hat{\beta}$ (33.5%) and the investment-share-weighted increase in the cost of capital (1.38%).[14]

Panel (a) in Figure J17 plots the calibrated shape of $\hat{q}(\boldsymbol{s})$ as a function of size-sorted cumulative employment share in a simulated panel of 1000 firms. The figure shows that the smallest firms experience an increase in their relative price of investment of around 10%, while the largest firms experience a decrease of around 5%.

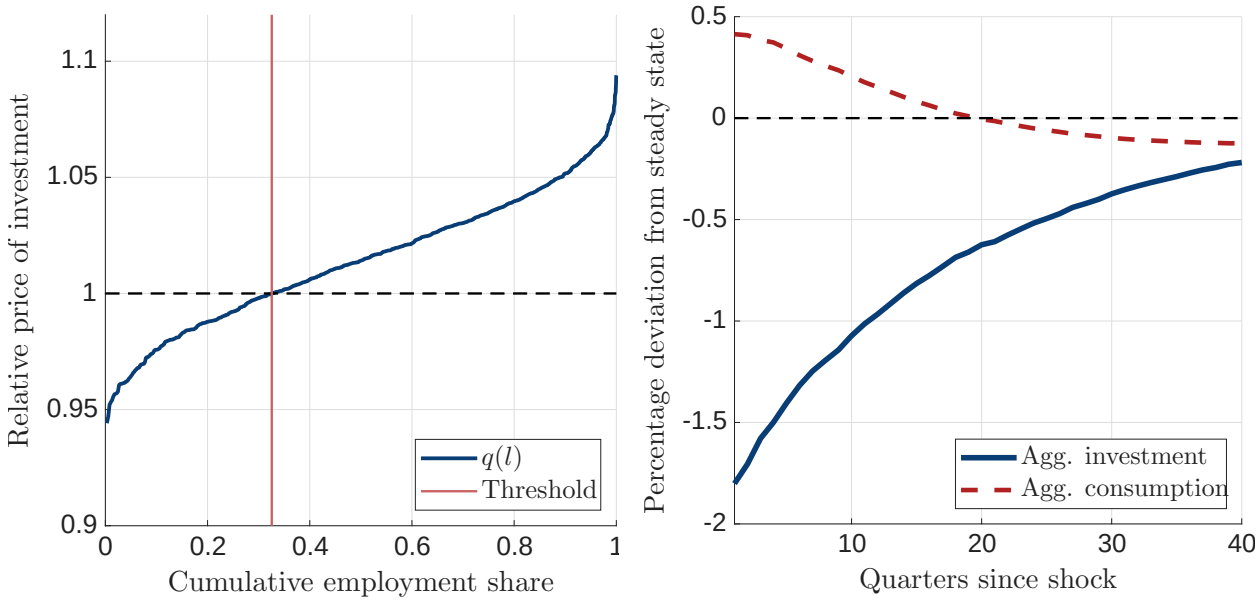
J.2 Results

Panel (b) of Figure J17 displays the impulse response to a one standard deviation shock to λ. On impact, aggregate investment contracts by 1.83%. This response operates almost entirely through the extensive margin, as the shock prevents formerly infra-marginal firms from undertaking planned investments.

The underlying mechanism is rooted in firm heterogeneity. In the model, the incentive to invest is strongest for firms with low initial capital relative to their optimal level. These are

[14] The investment-share-weighted increase in CAPM betas is 0.043 (see Section 8). The investment-share-weighted increase in the average WACC is around 0.21%, which corresponds to a 1.38% increase relative to the average hurdle rate of 15% in Gormsen & Huber (2025).

Figure J17: Distributions of Perc. Cost of Equity Shock and Impulse Responses of Investment



(a) Relative price of investment $\hat{q}(\boldsymbol{s})$ vs. size-sorted cumulative employment share in simulated panel

(b) Impulse Response of Aggregate Investment and Consumption after Shock to $\hat{q}(\boldsymbol{s})$

typically small, high-growth firms consistent with firm-level data. Because they are actively growing to reach their optimal scale, these small firms constitute the majority of infra-marginal investors in the economy.

The adverse shock λ to the relative price of capital lowers the net benefit of investing, pushing these formerly infra-marginal firms to be marginal and cancel their investment. Conversely, larger firms, who receive a subsidy, are less likely to investment since they are closer to their optimal scale, and their decisions are thus less affected. Therefore, the sharp drop in aggregate investment reflects the withdrawal of many small, marginal firms from investment.

K Data Appendix and Variables Definitions

Variable	Definitions and Source
Welch (2022) *Simply Better Market Beta*	Estimated using WLS regressions with exponentially decaying weights of 4-month half-life on an expanding window after winsorizing daily returns at −2x and +4x the contemporaneous market return. See Welch (2022) for estimation details.
Rolling-window CAPM Beta (21 days)	Month-end estimates using 21 daily returns (WRDS Beta Suite)
Rolling-window CAPM Beta (252 days)	Month-end estimates using 252 daily returns (WRDS Beta Suite)
Rolling-window CAPM Beta (156 weeks)	Month-end estimates using 156 weekly returns (WRDS Beta Suite)
Rolling-window CAPM Beta (36 months)	Month-end estimates using 36 monthly returns (WRDS Beta Suite)
Cash Flow Beta	Estimated as $\text{ROE}_{i,t} = \alpha_i + \beta_i^{CF}\text{ROE}_{Mkt,t} + \varepsilon_{i,t}$ where ROE is the ratio of clean surplus accounting $X_t = BE_t - BE_{t-1} + D_t$ to beginning-of-the-period book equity (BE_{t-1}). D_t are gross dividends computed as the difference between CRSP returns and returns excluding dividends (Cohen et al., 2009). Estimated quarterly from 1975 to 2018 separately for each firm in Compustat using an expanding window of observations.
Consumption Beta (Kim et al., 2024)	Weighted average of firms' and industry peers' consumption βs $\beta_{i,t}^{C\star} = \frac{2}{3}\beta_{i,t}^{C} + \frac{1}{3}\beta_{-i,t}^{C}$, normalized by standard deviation ($\sigma_{\beta_{i,t}^{C\star}}$). See Kim et al. (2024) for details on estimation.
Equity Ratio $\frac{E}{E+D}$	Using quarterly Compustat variables $\frac{cshoq \times prccq}{atq - ceqq - txdbq + cshoq \times prccq}$
Benchmarking Intensity (BMI)	See definition in Eq. (1). Provided by Pavlova & Sikorskaya (2023).
Institutional Ownership (IOR)	Estimated using Thomson-Reuters 13F filings on WRDS using the code provided by Palacios, Moussawi, Glushkov (2009).
Net Equity Issuance	Equity Issuance − Buy Backs over Total Assets Provided by Jensen et al. (2023)
Bid-Ask Spread	Provided by Abdi & Ranaldo (2017)
Amihud Illiquidity	Provided by Jensen et al. (2023)
Momentum	Cumulative return over the past 12 months Provided by Jensen et al. (2023)
May Market Capitalization Rank	Firm's market capitalization rank in May of each year calculated using methodology of Ben-David et al. (2019). See Appendix C.
Banding Variables/Controls	Indicator variables for having rank-date market cap in the reconstitution bands, an indicator for being in the Russell 2000 in May before reconstitution, and the interaction of these indicators. See Appendix C.

Variable	Description and Source
Average Implied Cost of Capital (ICC)	Average of the residual income models of Gebhardt et al. (2001) and Claus & Thomas (2001), and the dividend discount models of Easton (2004) and Ohlson & Juettner-Nauroth (2005). Provided by Eskildsen et al. (2024)
Managers' Perceived Cost of Capital	Provided by Gormsen & Huber (2025) For details see https://www.costofcapital.org/
Managers' Hurdle Rate	Provided by Gormsen & Huber (2025) For details see https://www.costofcapital.org/
Net Flows into Passive Mutual Funds and ETFs	Excluding feeder funds and funds of funds. Provided by Morningstar Direct.
Net Flows into Active Mutual Funds and ETFs	Excluding feeder funds and funds of funds. Provided by Morningstar Direct.
Total Net Assets of Passive Mutual Funds and ETFs	Excluding feeder funds and funds of funds. Provided by Morningstar Direct.
Total Net Assets of Active Mutual Funds and ETFs	Excluding feeder funds and funds of funds. Provided by Morningstar Direct.
Investment Rate	Using annual Compustat variables $\frac{capx_t}{ppegt_{t-1}}$
Tobin's q^{tot}	Total Tobin's q provided by Peters & Taylor (2017).
Current Ratio	Using annual Compustat variables $\frac{act_t}{lct_t}$
Leverage	Using annual Compustat variables $\frac{lt_t}{at_{t-1}}$
Cash Flow	Using annual Compustat variables $\frac{ib_t+dp_t}{ppegt_{t-1}}$
Operating Leverage	Using annual Compustat variables $\frac{xsga_t}{sale_t}$
Firm Age	Years since first appearance in Compustat

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